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Design and Deveopment of a pedagogically constrained Artificial intelligent-Based Educational Video Generator Using Retrieval Augmented Generation, Knowledge Graphs

Chapter one

introduction

1.1 Background of the study

Education is one of the most important drivers of human and societal development. Education helps individuals to acquire knowledge, skills, values, and competencies that enable them to improve their quality of life and contribute to economic and social progress. The world has become increasingly knowledge-driven, and with it, the demand for effective and accessible learning systems has grown. Consequently, improving the quality of teaching and learning has become a major concern for educational institutions worldwide (World Bank, 2018).

Educational practices have been significantly transformed by advances in information and communication technologies. Digital learning technologies have granted more access to education by enabling learners to get instructional materials anytime and from any location. Technologies such as online learning platforms, learning management systems, and mobile applications have changed the traditional classroom environment and created opportunities for more flexible and learner-centred education. These developments have made digital learning an essential component of modern educational systems (UNESCO, 2023).

There are several digital learning tools available, among which instructional videos have emerged as one of the most effective resources for teaching and learning. Video-based learning combines visual and auditory information, making abstract concepts

easier to understand and improving learner engagement and retention. Blended teaching, which combines the benefits of face-to-face instruction with the flexibility and accessibility of digital resources, has received special attention in recent times, which has led to its development, resulting in several models including flipped classroom, a method in which learners engage with instructional content independently before class and use classroom time for discussion and application (Saboowala and Mishra, 2021).

The growing demand for educational videos has, however, created challenges in content development. Producing high-quality instructional videos requires significant time, technical expertise, and financial resources. As a result, educators and institutions have been exploring other approaches to content creation. Recent advances in Generative Artificial Intelligence (AI) have made it possible to automatically generate text, images, speech, and videos, creating new opportunities for educational content production and delivery (Kasneci *et al*., 2023).

The integration of AI into education has given rise to AI-driven educational systems capable of generating instructional materials and supporting personalised learning experiences. Intelligent tutoring systems and adaptive learning platforms increasingly employ AI technologies to provide customised learning support and feedback to students. These technologies have the potential to reduce teachers' workload, improve educational accessibility, and facilitate the large-scale production of learning materials, particularly in environments facing shortages of educational resources and qualified teachers (UNESCO and International Task Force on Teachers for Education 2030, 2024).

Despite these benefits, the application of generative AI in education presents several challenges. AI systems are prone to generating inaccurate information, commonly referred to as AI hallucinations. They may also produce content that is inconsistent with curriculum requirements, lacks pedagogical structure, or contains cultural and contextual biases. Such deficiencies are likely to have a negative effect on learning outcomes and reduce the reliability of the generated educational resources. Therefore, ensuring content accuracy and subject matter relevance has become a rising concern in the use of AI for instructional content generation (Holmes *et al.*, 2019).

Beyond factual accuracy, the educational quality of AI-generated content depends largely on its pedagogical effectiveness. Educational materials should be designed to promote meaningful learning, encourage learner engagement, and support the gradual development of knowledge and skills. Learning theories such as Constructivism, Scaffolding, and Cognitive Load Theory emphasise the importance of structured instruction and learner-centred content design. However, many existing AI systems are primarily optimised for information generation rather than pedagogical quality assurance (Luckin and Cukurova, 2019).

To address these limitations, recent research has focused on integrating curriculum models, pedagogical constraints, and adaptive feedback mechanisms into AI-driven educational systems. These approaches seek to ensure that generated content aligns with curriculum objectives, follows logical instructional sequences, and supports effective learning processes. However, existing methods still provide limited guarantees regarding content accuracy, conceptual coherence, and subject matter relevance, particularly in the generation of educational videos (Li *et al.*, 2024).

The need to validate and optimise AI-generated instructional content has therefore become increasingly important. Mathematical modelling and validation frameworks provide a promising approach for evaluating the quality of generated educational materials by assessing factors such as accuracy, curriculum alignment, pedagogical quality, and relevance to the subject domain. Consequently, there is a need to develop

AI-based systems capable of generating educational videos that are not only technically accurate but also pedagogically sound, curriculum-aligned, and effective in promoting meaningful learning experiences.

1.2 Statement of the problem

Despite the increasing adoption of artificial intelligence for educational content generation, there have been significant challenges regarding the pedagogical quality of such content, especially for video content. While AI systems can efficiently produce large volumes of instructional materials, they are primarily designed for information generation rather than to support effective teaching and learning. This creates a situation in which the instructional value of AI-generated videos is not guaranteed, as they often prioritize information delivery over structured learning processes that support understanding (Wayne Holmes *et al*., 2019; Mutlu Cukurova and Rose Luckin, 2019). As a result, many AI-generated educational videos present content in a passive, pre-structured manner, often failing to align with established pedagogical principles such as Constructivism, Scaffolding, and Cognitive Load Theory. This misalignment can limit opportunities for active engagement, critical thinking, and deep understanding, potentially leading to cognitive disengagement and over-reliance on AI-generated outputs (Hai, Mai and Hanh, 2026). In addition, many conventional generative AI systems depend largely on static parametric knowledge acquired during training, which may result in outdated, context-insensitive, or pedagogically shallow instructional content. Emerging approaches such as Retrieval-Augmented Generation (RAG) and Knowledge Graphs offer potential solutions by enabling AI systems to retrieve contextually relevant information dynamically and organize concepts through structured semantic relationships, thereby improving the accuracy, coherence, contextual relevance, and instructional sequencing of AI-generated educational videos.

1.3 Aim and objectives

The aim of this study is to enhance the pedagogical quality of AI-generated educational videos by developing a structured pipeline that integrates Retrieval

Augmented Generation (RAG), knowledge graphs, and explicit pedagogical constraints to ensure effective instructional design and learner-centred content generation

The objectives are as follows

I. To design a knowledge graph-based representation of domain concepts that captures conceptual relationships and prerequisite structures for educational content and a retrieval-augmened generation (RAG) component that retrieves relevant instructional materials to support accurate and context aware content generation.
II. To develop a unified video production pipeline using that combines knowledge graphs and RAG and to integrate pedagogical constraints (such as scaffolding, cognitive load theory, and instructional sequencing) into the pipeline
III. To evaluate the effectiveness of the proposed system in terms of pedagogical quality, conceptual coherence, and learner comprehension compared to conventional AI-generated instructional videos and human-recorded videos

1.4 Justification for the study

This study is important because artificial intelligence is increasingly being used to generate instructional videos and other educational content. However, most generative AI systems are designed to produce information rather than support effective learning, often resulting in content that lacks proper instructional structure.

To address this challenge, the study combines Retrieval-Augmented Generation (RAG, a method for finding and using information from an external source), knowledge graphs (a way of organizing information so that data are connected like a network), and pedagogical rules to improve both the accuracy and educational quality of AI-generated videos. While RAG enhances content accuracy and relevance, knowledge graphs help organise concepts into meaningful learning sequences. Given the growing need for accessible and scalable learning resources, this research seeks to develop AI-

generated instructional videos that are accurate, well-structured, and effective for learning.

1.5 Scope and limitations of the study

This study focuses on the development and evaluation of an AI-based pipeline for generating instructional videos using a Large Language Model, Retrieval-Augmented Generation (RAG), and knowledge graphs. The system aims to produce factually accurate and pedagogically structured content that supports conceptual progression and effective learning. Evaluation will focus on instructional quality, including clarity, coherence, and alignment with learning objectives. However, the study is limited to selected subject domains and a controlled experimental setting. It does not consider real-time learner interaction or long-term learning outcomes, and computational constraints may limit the complexity and richness of the generated instructional videos.

The knowledge graph used in the system will be limited to selected subject domains and will not cover all areas of education. Furthermore, the system will not account for real-time learner interaction or adaptive personalization based on individual student performance; the evaluation will be limited to a short-term assessment of the quality of generated content rather than long-term learning outcomes.

1.6 Definition of terms

1. Artificial Intelligence (AI): The simulation of human intelligence in machines that are programmed to perform tasks such as learning, reasoning, and content generation.
2. Generative AI: A class of artificial intelligence models capable of producing new content such as text, images, audio, or video based on training data.

3. Hallucination (AI): A situation where an AI system generates incorrect, misleading, or completely fabricated information presented as factual.
4. Instructional Design: The systematic planning of educational content to ensure effective learning, often based on theories such as Constructivism and Cognitive Load Theory.
5. Pedagogical Alignment: The degree to which educational content follows established teaching and learning principles to support understanding and learner engagement.
6. Constructivism: A learning theory that emphasizes learners actively constructing knowledge through experience, interaction, and reflection.
7. Scaffolding: A teaching method where learners are gradually guided through learning tasks, with support reduced as competence increases.
8. Cognitive Load Theory: A theory that explains how learning is affected by the amount of information processed in working memory, emphasizing the need to avoid overload.
9. Retrieval-Augmented Generation (RAG): An AI technique that improves content generation by retrieving relevant external information and combining it with language model outputs.
10. Knowledge Graph: A structured representation of information showing relationships between concepts, used to organize and connect domain knowledge.
11. Conceptual Coherence: The logical and consistent flow of ideas in instructional content, ensuring concepts are connected meaningfully.
12. Instructional Video: A video designed to teach specific knowledge or skills, often used in educational settings for blended or online learning.
13. AI-Generated Content: Educational or informational material created using artificial intelligence systems without direct human authorship of the content structure.
14. Over-reliance (on AI): Excessive dependence on AI tools, which may reduce critical thinking and independent problem-solving skills in learners.

15. Cognitive Engagement

    The level of mental effort and active thinking a learner applies when interacting with learning material.

16. Learning Objectives: Clear statements that define what learners are expected to know or be able to do after instruction.

17. Metacognition: The awareness and regulation of one's own thinking and learning processes.

18. Instructional Sequencing: The logical ordering of content from simple to complex to support progressive learning.

19. Human-in-the-loop (HITL): A system design where human input is used to guide, review, or correct AI-generated outputs.

20. Adaptive Learning: A learning approach where content or instruction is adjusted based on the learner's performance or needs.

21. Large Language Model (LLM): An AI model trained on large text datasets to understand and generate human-like language.

22. Educational Technology (EdTech): The use of digital tools and systems to support teaching, learning, and educational management.

CHAPTER TWO

LITERATURE REVIEW

2.1 Foundations of learning

2.1.1 Definition of learning

Learning is a continuous process through which individuals acquire knowledge, skills, values, and competencies. It involves not just the absorption of information, but also the ability to understand, apply, and adapt that knowledge in different contexts. In educational psychology, learning is often described as a relatively permanent change in behavior or mental processes resulting from experience or practice (Illeris, 2018).

Learning is not purely cognitive; it also involves emotional and social dimensions. For example, a learner's motivation, interest, and interaction with others can significantly influence how well they understand and retain information. In structured educational settings, learning is guided by clearly defined objectives, instructional strategies, and assessment methods. As education continues to evolve, especially with the integration of digital technologies, the focus has shifted toward making learning more engaging, flexible, and learner-centered. However, regardless of the tools used, effective learning still depends on how well content is designed to align with how people naturally process information.

2.1.2 Theories of learning

Over time, different theories have been developed to explain how learning occurs, with the most prominent being behaviorism, cognitivism, and constructivism.

Behaviorism focuses on observable changes in behavior and emphasizes the role of external stimuli in shaping learning. According to Skinner (1953), learning occurs through reinforcement, where desired behaviors are strengthened through rewards or

repetition. This approach has influenced traditional teaching methods, particularly those that rely on drills, practice, and direct instruction.

Cognitivism shifts the focus from external behavior to internal mental processes. It views learners as active participants who process, store, and retrieve information. Piaget (1970) explained that learning occurs through processes such as assimilation and accommodation, where new information is integrated into existing mental structures. This theory highlights the importance of organizing content in a way that supports understanding and memory.

Constructivism takes this a step further by suggesting that learners actively construct their own knowledge through experience and interaction. Vygotsky (1978) emphasized the social nature of learning, arguing that interaction with others plays a crucial role in cognitive development. Concepts such as collaboration, guided learning, and scaffolding are rooted in this theory.

In practice, modern educational systems often combine elements from all three theories to create more effective learning experiences.

2.1.3 Cognitive processes in learning

Learning is deeply connected to several cognitive processes, including attention, perception, memory, and problem-solving. These processes determine how information is received, interpreted, stored, and later retrieved.

Attention is the first step in learning, as it determines what information a learner focuses on. Without attention, information is unlikely to be processed further. Once attention is captured, perception helps the learner interpret the information based on prior knowledge and experiences. Memory then plays a crucial role, as it allows information to be stored and retrieved when needed. According to Mayer (2021), learning is most effective when instructional materials are designed in a way that supports these cognitive processes.

One important concept in this area is cognitive load, which refers to the amount of mental effort required to process information. If the content is too complex or poorly organized, it can overwhelm the learner and reduce understanding. This is especially important in multimedia learning environments, where text, audio, and visuals must be carefully balanced to avoid overload.

Another key factor is prior knowledge. Learners do not approach new information as blank slates; instead, they build on what they already know. This means that instructional content should be structured in a way that connects new concepts to existing knowledge, making learning more meaningful and easier to retain.

2.1.4 Factors affecting learning effectiveness

Learning effectiveness is influenced by a combination of learner-related, instructional, and environmental factors.

Learner-related factors include motivation, prior knowledge, cognitive ability, and learning preferences. Motivation, in particular, plays a critical role, as learners who are interested and engaged are more likely to actively participate in the learning process and achieve better outcomes.

Instructional design is another major factor. The way content is structured, presented, and delivered can significantly impact understanding. Clear learning objectives, logical sequencing, and the use of examples and multimedia elements can enhance comprehension (Mayer, 2021). Poorly designed materials, on the other hand, can confuse learners and reduce effectiveness.

The learning environment also matters. This includes both physical settings, such as classrooms, and digital environments, such as online learning platforms. A supportive and interactive environment encourages engagement, while a poorly designed one can lead to distraction and cognitive overload.

Finally, technological factors are becoming increasingly important. Digital tools such as learning management systems, video platforms, and AI-based systems have transformed how learning takes place. These technologies offer flexibility and scalability, however, their effectiveness depends on how well they are aligned with pedagogical principles (Yıldızhan Bora & sahin Kolemen, 2025). When used appropriately, they can significantly enhance learning outcomes; however, if poorly implemented, they may overwhelm learners or reduce engagement.

2.1.5 Digital technologies for learning

learning technologies have evolved from traditional classroom-based approaches to more interactive and technology-driven systems.

Traditionally, learning has been facilitated through methods such as lecture-based teaching, demonstrations, and guided instruction. Lecture-based learning remains one of the most widely used approaches, particularly in higher education. However, studies have shown that this method often promotes passive learning and may limit the development of critical thinking and problem-solving skills (Rani, 2020). As a result, more learner-centered approaches have gained attention.

One such approach is problem-based learning (PBL), where learners engage with real-world problems to develop critical thinking and practical skills. Similarly, the flipped classroom model allows students to study instructional materials, often in the form of videos, before class, while classroom time is used for discussion and application. This approach has been shown to improve student engagement and learning outcomes (Ullah & Jinah, 2023; Li *et al.*, 2022).

In addition to these pedagogical strategies, simulation-based learning and collaborative learning have also become important tools, particularly in fields that require practical application of knowledge. These approaches emphasize active participation, interaction, and experiential learning, which are key components of effective education.

With the advancement of technology, digital learning tsechnologies have become increasingly prominent. These include learning management systems (LMS), virtual classrooms, and multimedia platforms that support flexible and scalable learning environments. Digital tools enable remote learning, self-paced study, and access to a wide range of educational resources (Yıldızhan *et al.,* 2025).

Among these tools, video-based learning has emerged as a very important medium. Educational videos are widely used in flipped classrooms and online learning environments due to their ability to present information in a clear and engaging manner. However, the production of high-quality educational videos remains resource-intensive, requiring time, technical expertise, and instructional design knowledge.

This challenge has led to increasing interest in automated and AI-driven learning tools, particularly those that can support the efficient creation of educational content. As a result, emerging technologies such as generative AI, retrieval-augmented systems, and knowledge-based frameworks are being explored to improve the scalability and effectiveness of educational video production.

2.2 Multimedia learning and educational videos

2.2.1 Concept of multimedia learning

One pronounced digital learning technology is multimedia learning which refers to the use of multiple forms of content such as text, images, audio, animation, and video to support the learning process. Instead of relying on a single mode of instruction, multimedia learning combines different sensory inputs to enhance understanding and retention. This approach is based on the idea that learners process information more effectively when it is presented in both visual and verbal forms.

In modern education, multimedia learning has become increasingly important due to the growth of digital technologies. Learning platforms now integrate videos, graphics,

and interactive elements to create more engaging and flexible learning experiences. These tools allow learners to access content at their own pace and revisit materials as needed, which supports deeper understanding. However, the effectiveness of multimedia learning depends largely on how well the content is designed and structured.

2.2.2 Cognitive theory of multimedia learning

The Cognitive Theory of Multimedia Learning, developed by Mayer (2021), provides a framework for understanding how multimedia content supports learning. The theory is based on three main assumptions: learners have separate channels for processing visual and auditory information, each channel has a limited capacity, and learning occurs when learners actively process information.

According to this theory, effective multimedia instruction should be designed to reduce unnecessary cognitive load and focus on essential information. For example, combining relevant visuals with clear narration can improve understanding, while excessive or poorly organized content can overwhelm the learner. Principles such as coherence (removing irrelevant information), signaling (highlighting key points), and redundancy (avoiding unnecessary repetition) are important in designing effective multimedia materials.

2.2.3 Role of videos in education

Video has become one of the most widely used tools in modern education. It plays a central role in online learning, flipped classrooms, and digital learning environments. Videos allow complex concepts to be explained in a visual and dynamic manner, making them easier to understand compared to text-based materials alone.

Research shows that video-based learning can improve student engagement, motivation, and knowledge retention when properly designed (Morgado *et al.*, 2024). Videos also support self-paced learning, as learners can pause, rewind, and review

content as needed. This flexibility is particularly useful in diverse learning environments where students have different learning speeds and preferences.

In flipped classroom models, videos are often used to deliver foundational knowledge before class, allowing classroom time to be used for discussion and problem-solving (Li *et al.*, 2022). This highlights the growing importance of video as a core instructional tool. However, the increasing reliance on video also raises challenges related to content quality, production cost, and scalability.

2.2.4 Types of educational videos

Educational videos can be categorized based on their purpose, design, and instructional approach. Some of the most common types include:

I. Lecture-based videos: These involve recorded classroom lectures or direct instruction by an instructor. They are simple to produce but may be less engaging if not designed properly.
II. Animated explainer videos: These use animations and graphics to simplify complex concepts, making them particularly effective for abstract or technical topics.
III. Demonstration videos: These show step-by-step processes or procedures, commonly used in science, engineering, and vocational training.
IV. Simulation videos: These replicate real-world scenarios, allowing learners to observe processes that may be difficult to experience directly.
V. Interactive videos: These include embedded questions, quizzes, or decision points that actively engage the learner during the video.

Each type serves a specific instructional purpose, and the choice of video format should align with the learning objectives and learners' needs.

2.2.5 Benefits and limitations of video-based learning

Video-based learning offers several advantages. First, it enhances engagement by combining visual and auditory elements, which can make learning more interesting and accessible. Second, it supports better understanding of complex concepts through visualization. Third, it allows for flexible and self-paced learning, enabling learners to control their learning experience.

Additionally, videos can be reused and distributed widely, making them a scalable solution for education. This is particularly important in online and distance learning environments.

Despite these benefits, video-based learning also has limitations. One major challenge is the time and effort required for producing high-quality educational videos. The process often involves scripting, recording, editing, and instructional design, which can be time-consuming (Zawacki-Richter *et al.,* 2019). Poorly designed videos can also lead to cognitive overload, reducing their effectiveness.

Another limitation is the lack of interactivity in many video formats, which can result in passive learning. Furthermore, videos may not fully address individual learning differences unless they are integrated into adaptive or interactive systems.

These limitations highlight the need for more efficient and intelligent approaches to educational video production. Emerging technologies, particularly artificial intelligence, offer potential solutions by automating aspects of video creation while maintaining instructional quality.

2.3 Instructional design for educational content

2.3.1 Principles of instructional design

Instructional design refers to the systematic process of creating educational experiences that facilitate effective learning. It involves analyzing learner needs, defining learning objectives, structuring content, and selecting appropriate delivery

methods. The goal is to ensure that instructional materials are clear, engaging, and aligned with desired learning outcomes.

One widely used framework is the ADDIE model, which consists of Analysis, Design, Development, Implementation, and Evaluation. This model emphasizes a structured approach to content creation, ensuring that each stage of instruction is carefully planned and assessed (Branch, 2009). Another important concept is constructive alignment, which ensures that learning objectives, instructional activities, and assessment methods are all aligned to support meaningful learning (Biggs, 1996).

Effective instructional design also considers cognitive principles, such as managing cognitive load and organizing information in a way that supports comprehension. According to Mayer (2021), well-designed instructional materials should guide learners' attention, reduce unnecessary complexity, and promote active processing of information. These principles are especially important in multimedia environments, where poorly structured content can overwhelm learners.

2.3.2 Designing effective educational videos

Designing effective educational videos requires more than simply information; it involves applying instructional design principles to ensure that the content supports learning. Educational videos should be structured in a clear and logical sequence, beginning with learning objectives, followed by explanation, examples, and summary.

One important consideration is segmentation, where content is broken into smaller, manageable parts. This helps learners process information more effectively and reduces cognitive overload. Additionally, the use of visuals and narration should be carefully coordinated to reinforce key concepts rather than distract from them (Mayer, 2021).

Clarity and simplicity are also essential. Overly complex visuals, excessive text, or irrelevant information can reduce the effectiveness of a video. Instead, instructional

videos should focus on key ideas and use visual cues, such as highlights or animations, to guide attention.

Another critical aspect is alignment with learning objectives. Every element of the video, whether visual, auditory, or textual should contribute to achieving the intended learning outcome. This ensures that the video is not only engaging but also educationally meaningful.

2.3.3 Challenges in traditional educational content production

Despite the importance of instructional design, traditional approaches to educational content production face several challenges. One of the main issues is the time and effort required to develop high quality instructional materials. Creating educational videos, for example, involves multiple stages, including planning, scripting, recording, editing, and evaluation (Mayer, 2021). This process often requires collaboration between subject matter experts, instructional designers, and technical specialists, making it resource-intensive (Dick, Carey & Carey, 2015) . As a result, many educational institutions struggle to produce large volumes of high quality content.

Another challenge is maintaining consistency and quality across different materials. Without standardized processes, instructional content may vary in structure, clarity, and effectiveness. This can negatively impact learning outcomes (Morrison, Ross & Kemp, 2019).

Additionally, traditional content production methods are not easily scalable. Updating or modifying content can be time-consuming, especially when dealing with rapidly changing subject areas. This limits the ability of educators to provide up-to-date and personalized learning materials (Hodges *et al.*, 2020) .

These challenges highlight the need for more efficient and flexible approaches to instructional content creation. Advances in technology, particularly in artificial intelligence, offer promising solutions by enabling automation and improving the scalability of educational content production (Luckin *et al.*, 2016) .

Literature shows a clear progression inattempts to address these production challenges over the past decade. Early work documented the scale of the problem: Hollands and Tirthali (2014) estimated massive open online course (MOOC) production costs at $39,000–$325,000, with video accounting for a large share, while Guo, Kim, and Rubin (2014) empirically examined how production decisions such as video style and length affected student engagement, offering some of the first evidence-based guidance for effective rather than merely expensive instructional video.

By the mid-2010s, researchers began targeting the human-effort problem directly, experimenting with low-cost production schemas, repurposed video, and instructor self-production to cut costs without sacrificing usability. A major shift followed from speech and vision synthesis, (Jia *et al.,* 2018) demonstrated transfer learning for multispeaker text-to-speech synthesis, a technique that later underpinned fully automated systems such as Wang, Song, and Jha's (2022) AutoLV, which generated complete lecture videos from slides, a reference voice, and a reference portrait — explicitly designed to reduce instructor workload and allow videos to be updated without recapturing footage.

Most recently, research has moved toward multi-agent, end-to-end pipelines. Wang *et al*. (2025) introduced PaperTalker, which coordinates slide generation, subtitling, cursor grounding, and talking-head synthesis in parallel, achieving over a 6x speedup. Around the same time, Netland *et al*. (2025) found AI-generated videos took just 3 to 4 hours to produce versus 12 to 24 hours for human-made ones, with no significant difference in learning outcomes, closing the loop between the cost problem identified a decade earlier and the automated solutions now available.

2.4 Artificial intelligence in education

2.4.1 Overview of artificial intelligence

Artificial Intelligence (AI) refers to the development of computational systems capable of performing tasks that normally require human intelligence, such as reasoning, learning, perception, and decision-making (Russell & Norvig, 2021). These systems are designed to analyze data, detect patterns, and generate outputs that simulate aspects of human cognition (Kaplan & Haenlein, 2019).

In modern applications, AI is no longer limited to theoretical research but is actively used across domains such as healthcare, transportation, and education (Holmes *et al.*, 2019). In education, AI supports teaching and learning by automating tasks, improving accessibility, and enabling adaptive learning experiences (Zawacki-Richter *et al.*, 2019).

AI-based systems are particularly valuable in environments where large volumes of data must be processed to personalize learning experiences. These systems can recommend learning paths, generate content, and provide feedback based on learner interaction patterns (Luckin *et al.*, 2016).

However, despite these advantages, the effectiveness of AI in education depends heavily on how well it aligns with pedagogical principles and instructional design strategies (Williamson & Eynon, 2020).

2.4.2 Evolution of AI in education

The use of AI in education has evolved significantly from simple rule-based systems to complex machine learning models. Early educational AI systems, often referred to as Intelligent Tutoring Systems (ITS), relied on predefined rules to guide learner interaction (VanLehn, 2011). These systems were limited in flexibility and could not adapt well to diverse learner needs (Holmes *et al.*, 2019).

A shift began in the mid-2000s with the emergence of educational data mining (EDM) and learning analytics, as institutions accumulated enough learner-interaction data (from learning management systems and, later, MOOCs) to train statistical and

machine learning models rather than hand-coded rules. This allowed systems to move from fixed instructional logic toward models that could adapt based on observed learner behavior, enabling more personalized feedback and dynamic adjustment of instructional content (Ali, 2026).

More recently, deep learning and large-scale models have further advanced AI capabilities, enabling systems to generate human-like text, summarize information, and assist in content creation (Brown *et al.*, 2020). These advancements have expanded AI's role in education beyond tutoring into content generation and instructional design (Zawacki-Richter *et al.*, 2019).

2.4.3 Applications of AI in teaching and learning

The applications of artificial intelligence in education have followed distinct but converging trajectories. Automated assessment has the longest history: Page (1966) introduced Project Essay Grader, the first automated essay scoring system, later refined by systems using latent semantic analysis (Foltz *et al.*, 1999) and NLP-based scoring (Burstein *et al.*, 1998), consolidating into the mature grading and feedback systems described by Shermis and Burstein (2013), which reduced educator workload while accelerating feedback cycles (Zawacki-Richter *et al.*, 2019).

Adaptive learning systems built on decades of mastery-learning research (Kulik, Kulik & Bangert-Drowns, 1990), culminating in Kulik and Fletcher's (2016) meta-analysis of 50 studies, which found intelligent tutoring systems raised test scores by 0.66 standard deviations over conventional instruction, strong evidence for dynamically adjusting content difficulty to learner progress.

AI-powered chatbots emerged more recently as conversational AI matured: Winkler and Söllner (2018) documented pedagogical chatbots extending learner support beyond the classroom with real-time guidance.

Most recently, generative content creation has built on large language models such as GPT-3 (Brown *et al*., 2020) to produce instructional text, images, and video at scale, though Nguyen *et al*. (2023) caution that accuracy and pedagogical alignment remain unresolved concerns.

2.4.4 Benefits and challenges of AI in education

As these applications matured, research attention shifted from documenting the benefits of artificial intelligence to scrutinizing the risks that came with them. Personalization is the most well-evidenced benefit: Pane *et al*. (2017), in a large research and development (RAND) study of 62 U.S. schools, found real achievement gains from tailoring instruction to individual learners, though effects varied by context (Holmes *et al*., 2019). This supported the efficiency and scalability arguments made by Zawacki-Richter *et al*. (2019) and Luckin *et al*. (2016), who noted that automating grading and content generation frees educators to focus on teaching rather than routine tasks (Ali, 2026).

But the same capabilities driving these benefits also introduced new concerns. The accuracy issues already flagged in generative content (Brown *et al*., 2020) point to a broader reliability problem, compounded by the risk that AI systems reproduce biases in their training data (Williamson & Eynon, 2020).

2.5 Natural language processing and language models

2.5.1 Overview of natural language processing

Natural language processing (NLP), a subfield of Artificial intelligence concerned with enabling computers to understand, interpret, and generate human language in a meaningful way (Jurafsky & Martin, 2023), has contributed massively to the growing usefulness of AI in education content creation. In its earliest form, NLP relied on rule-based and symbolic methods that could handle narrow, well-defined language tasks but broke down outside those constraints (Jurafsky & Martin, 2023). Subsequent development followed a consistent pattern of trading rigidity for statistical learning from data: by the 2000s, n-gram approaches used fixed word sequences to model language statistically, though they remained limited by local context (Manning *et al*., 2008), and in 2003, neural language models introduced distributed word representations that captured semantic relationships more effectively than these statistical predecessors (Bengio *et al*., 2003). Recent advancement has been geared toward scale and context: since 2017, transformer-based architectures have processed entire sequences simultaneously (Vaswani *et al*., 2017), and by 2020 this had produced large language models capable of coherent, context-aware generation at a scale earlier approaches could not achieve (Brown *et al*., 2020).

This trajectory from rigid rules, to statistical pattern-matching in the 2000s, to context-aware generation at scale by 2020 — has led some scholars to anticipate continued gains in NLP capability as models and datasets grow (Goldberg, 2017). However, scaling language models further is unlikely to resolve a more fundamental challenge: NLP systems still struggle with deep semantic understanding in tasks that require structured knowledge and contextual reasoning (Bender *et al*., 2021). For educational video content specifically, this matters directly, generating accurate, pedagogically sound video material depends on more than fluent language generation; it depends on grounded, structured understanding of subject matter, which is exactly what current systems have not solved. It is this persistent gap between what NLP can generate

fluently and what it can generate correctly and pedagogically for video-based instruction — that this project addresses.

2.5.2 Language models and their types

Language models are a core component of NLP systems and are designed to predict or generate sequences of words based on learned patterns from data. These models estimate the probability of a word or sequence of words occurring in a given context (Jurafsky & Martin, 2023).

Early language models which surfaced in the 2000s were based on statistical approaches such as n-grams, which relied on fixed word sequences and had limited ability to capture long-range dependencies in text (Manning *et al.*, 2008). While simple, these models were not effective for complex language tasks due to their reliance on local context only.

With the development of neural networks in 2003, Neural Language Models (NLMs) were introduced, allowing systems to learn distributed representations of words and capture semantic relationships more effectively (Bengio *et al.*, 2003). This marked a significant improvement over traditional statistical models, models could now represent that “photosynthesis” and “chlorophyll” were related concepts, not just adjacent tokens, but NLMs still processed text in comparatively short, fixed windows, so they could not sustain the coherence needed across an entire explanation or lesson.

By 2017, Transformer-based models had became the dominant architecture in NLP. Models such as BERT and GPT used attention mechanisms to process entire sequences of text simultaneously, allowing them to capture long-range dependencies more effectively (Vaswani *et al.*, 2017). These models formed the foundation of modern Large Language Models (LLMs), which are capable of generating coherent and context-aware text at scale (Brown *et al.*, 2020).

In educational contexts, these models are increasingly used for content generation, summarization, and conversational tutoring systems. However, their effectiveness depends on how well they are grounded in accurate and structured knowledge sources.

2.5.3 Limitations of language models

This dependency on grounding is precisely where language models fall short. Despite the architectural gains traced above, none were built to verify what they generate against a source of truth — each solved fluency, not accuracy. The clearest evidence is hallucination: the generation of plausible-sounding but factually incorrect information, since models are trained to predict text patterns rather than verify facts (Ji *et al.*, 2023). This is not a flaw specific to one architecture — it follows from what every generation of language model, including the transformer-based LLMs in 2.5.2, is fundamentally designed to do.

Models also lack genuine contextual understanding: they process text through statistical association rather than human-like comprehension (Bender & Koller, 2020), producing outputs that are coherent in form but not necessarily accurate or logically consistent. They struggle with long-term reasoning and multi-step problem solving, especially where information must be retrieved from external or structured sources (Bommasani *et al.*, 2021), a gap that persists even in transformers, which solved coherence but not multi-step reasoning. Sensitivity to prompt phrasing adds further inconsistency, with different inputs producing different outputs for the same underlying request (OpenAI, 2023).

For education, where accuracy and logical progression are essential, these limitations strike at exactly what instructional content must have. Scaling models further is unlikely to help, since the problem is not insufficient fluency but the absence of any grounding mechanism which is what motivates the hybrid approaches, RAG and knowledge graph integration, taken up next.

2.6 Generative Artificial intelligence (Gen AI)

Generative AI, a class of artificial intelligence systems that are capable of creating new content rather than simply analyzing existing data extends language models from prediction to production. Instead of analysing existing data, these systems learn the underlying distribution of training data and generate new outputs that are statistically similar but not identical to it (Goodfellow *et al*., 2014). Its development follows its own trajectory, running alongside the progression from n-gram models to transformer-based LLMs described above.

In 2014, Generative Adversarial Networks (GANs) introduced the first influential approach, a generator network competing against a discriminator that evaluates authenticity, with the generator learning to produce increasingly convincing outputs as it is trained against the discriminator's feedback (Goodfellow *et al*., 2014). GANs were however built primarily for image synthesis, not language or instructional content. That same year, Variational Autoencoders (VAEs) offered an alternative approach, learning compressed latent representations of data from which new samples could be reconstructed. This was also applied mainly to image and representation-learning tasks (Kingma & Welling, 2014), again, general-purpose, not education-driven.

Neither GANs nor VAEs solved the coherence problem that limited earlier language models: sustaining an extended, contextually consistent explanation. That required the transformer architecture, which by 2017 could process entire sequences simultaneously through attention (Vaswani *et al.*, 2017), and which by 2020 had scaled into large language models such as GPT-3, capable of coherent, context-aware text generation (Brown *et al.*, 2020). Also around 2020, diffusion models emerged as a distinct approach for high-quality image and video generation (Ho *et al.*, 2020), the first generative architecture with a direct, structural relevance to educational video production specifically, rather than text alone.

Unlike GANs and VAEs, transformer- and diffusion-based generation has been explicitly applied to education, used to automate parts of the video production pipeline which include script writing, narration, and visual generation, reducing the time and expertise required (Pellas, 2025), with AI-generated instructional videos showing learning outcomes comparable to human-made content in some self-paced contexts (Leiker *et al.*, 2023), and supporting personalisation at a scale difficult to achieve manually (Holmes *et al.*, 2019). But this progression, like the language models before it, solves fluency and modality rather than accuracy, effectiveness still depends on prompt quality, training data, and instructional design constraints applied during generation, none of which current systems enforce automatically.

This dependency produces four recurring challenges:

I. factual inaccuracy and hallucination, the same problem where models generate plausible-sounding but factually incorrect information because they are trained to predict text patterns rather than verify facts. This is especially risky since learners may treat generated content as authoritative.

II. weak instructional alignment, since generative models are not inherently designed around pedagogical principles or curriculum standards (Kasneci *et al.*, 2023)

III. reproduction of training-data bias (Bender *et al*., 2021); and

IV. limited explainability, since it is often difficult to trace how a model arrived at a given output. Together, these point to the same missing ingredient already identified: an external, structured, verifiable source of truth the generative process can be checked against.

2.6.1 Advancements in Generative AI

The challenges hindering wide adoption of generative AI for educational content have over the years received attention, the earliest of these efforts predates modern generative AI entirely. Zhang, Lemoine, and Mitchell (2018) worked on reducing bias in model outputs, training a model to maximise predictive accuracy while simultaneously training an adversary network to strip out its ability to predict a protected attribute from those outputs. The result was adversarial debiasing, a working method, but one built for classification tasks, years before generative text and image models existed at scale.

Lewis *et al*. (2020) considered the accuracy problem from a different angle entirely: rather than adjusting a model's internal parameters, they proposed grounding a model's outputs in documents retrieved from an external knowledge base at generation time. The result was Retrieval-Augmented Generation (RAG), the first approach shown to measurably reduce hallucination by constraining generation to retrieved evidence, rather than relying on the model's memorised training data alone.

The following year, Bommasani *et al*. (2021) turned to a different challenge, explainability, surveying what was then a new class of large-scale "foundation models" and documenting how difficult it remained to trace a model's output back to a specific cause within it. Their conclusion was not a fix, but a formal acknowledgment that interpretability at this scale was, and remained, unsolved.

By 2023, enough scattered mitigation work had accumulated that two researchers moved to organise it. Ji *et al*. (2023) worked on hallucination specifically, reviewing

existing efforts and sorting them into three categories by where they intervene in the pipeline, the training data itself, the model's training process, or its output at inference time. In the same year, Ferrara (2023) worked on bias specifically, proposing a parallel three-part structure, pre-processing, in-processing, and post-processing interventions, giving the field, for the first time, a shared vocabulary for comparing bias-mitigation techniques rather than treating each new method as a one-off.

Gallegos *et al*. (2024) extended this bias work specifically to large language models, the architecture class that had by then become dominant, and concluded that despite years of accumulated technique, bias mitigation remained an open, unresolved problem rather than a solved one. That same year, the first education-specific effort in this timeline appeared: Jia *et al*. (2024) fine-tuned a large language model using Low-Rank Adaptation to align its outputs with Gagné's Nine Events of Instruction, a named pedagogical framework, directly targeting instructional alignment rather than accuracy, bias, or explainability.

2.6.3 Applications in educational content and video production

Early work on generative AI in education focused on the promise of scale rather than on testing it directly. Holmes *et al*. (2019) argued that generative systems could adapt content to individual learners' needs, preferences, and difficulty levels, framing personalisation as the field's central opportunity, while also flagging that data-driven personalisation raises privacy concerns. Bender *et al*. (2021) then showed that models trained on uncurated web corpora reproduce the biases in that data, meaning generated content risks carrying those biases into the classroom. Kasneci *et al*. (2023) synthesised this into a broader picture of the field, documenting automatic generation of explanations, summaries, and practice questions as an established application, while noting that generative models are not built with pedagogical principles in mind and so may not align with curriculum standards.

From 2023 onward, researchers moved from proposing and surveying these applications to actually building and testing them, particularly in instructional video. Leiker *et al*. (2023) ran the earliest of these trials, comparing AI-generated instructional videos against traditional human-created content and finding comparable learning outcomes in self-paced settings. Netland *et al*. (2025) scaled this test considerably: using generative tools including an LLM for scripting and avatar-based video creation, they produced four short teaching videos for a management course and tested them against human-made equivalents in an online experiment with 447 participants and 1,788 video-treatment observations. Learning outcomes measured through test performance were equivalent across conditions, though participants reported a slightly more favourable subjective experience with human-instructor videos.

Xu *et al*. (2025) tested a related but distinct question, whether a full AI-generated lecture pipeline, including synthesised virtual teacher characters and voices, could match recorded human lectures. They built and implemented an AI-generated lecture video pipeline specifically to evaluate learning performance and learner experience, and to test how synthesised virtual teacher characters, voices, and communication styles compare to human instructors, working from the hypothesis that AI teachers would produce lower social presence and trust than human ones despite closely resembling them. Struger *et al*. (2025) narrowed further, testing not learning outcomes but emotional response: in a mixed-method study, 55 participants viewed AI-avatar videos alongside videos with real instructors, with emotional engagement measured using FaceReader alongside quiz questions and interviews. They found AI avatars effectively conveyed content and elicited recall rates and positive emotional responses comparable to real instructors, though concerns remained around emotional authenticity and engagement.

Other trials tested production efficiency rather than pedagogy. The AutoLectures system, for instance, demonstrated conversion of static slides into narrated videos with synchronised highlights at a cost of under $1 per hour, showing that the automation

gains Pellas (2025) later described conceptually had already been demonstrated in a working system. A separate study with adult learners tested content type against outcome directly: in an experiment with 83 adult learners aged 18 to 64, researchers compared traditional instructor-led videos against AI-generated synthetic videos and found the AI-generated videos effective, though they noted that content accuracy and reliability remained a concern, which they mitigated in the study by using expert-curated source material.

129 Generative AI has a wide range of applications in education, particularly in content creation and instructional design. One of its key applications is the automatic generation of learning materials, including explanations, summaries, and practice questions, which can support both teachers and learners (Kasneci *et al*., 2023).

170 In video-based learning, generative AI is increasingly used to automate parts of the production pipeline, including script writing, narration, and visual generation. This reduces the time and technical expertise required to produce educational videos (Pellas, 2025).

AI-generated instructional videos have also been shown to support learning outcomes comparable to traditional human-created content in some contexts, particularly in self-paced learning environments (Leiker *et al*., 2023).

Additionally, generative AI enables personalized learning experiences, where content can be adapted to individual learner needs, preferences, and difficulty levels. This is particularly useful in large-scale online education systems where individualized instruction is difficult to achieve manually (Holmes *et al*., 2019).

However, while generative AI improves scalability, its effectiveness depends on the quality of prompts, training data, and instructional design constraints applied during content generation.

2.6.4 Challenges of generative AI in education

The trials reviewed above point to a consistent pattern, generative AI can match human produced content on measurable learning outcomes, but the harder challenges lie in bias, privacy and explainability. factual inaccuracy or hallucination remains one of the most critical issues, and it is particularly problematic in education, where accuracy is essential for learning.

Bender *et al*. (2021) argued bias would carry over from training data, a systematic review of AI-generated educational images found recurring gender, racial, cultural, age, and disability biases, alongside tension between visually appealing output and factual accuracy. A study of AI-generated religious education content found generative AI does not merely reflect but actively amplifies cognitive biases already present in training data. And a Global South STEAM lesson-planning study found AI-generated plans often lacked context sensitivity, frequently omitting cultural and linguistic features essential to equitable education. This gives Gallegos *et al*.'s (2024) conclusion, that bias mitigation remains unresolved, direct empirical support.

Bommasani *et al*.'s (2021) reported that interpretability remains unresolved. According to them This challenge has the strongest empirical grounding, and it explains the recurring finding that AI content matches test scores but lags on trust. A study of 41 chemistry teachers using an AI recommendation tool found explainability drives trust, but general AI knowledge alone does not translate into teachers trusting AI-EdTech. A review of 35 XAI-in-education studies similarly found methods like SHAP and LIME help students and educators understand outputs, with downstream effects on trust, motivation, and success.

Another challenge is the lack of instructional alignment. Generative models are not inherently designed with pedagogical principles in mind, which means the content they produce may not always align with learning objectives or curriculum standards (Kasneci *et al*., 2023).

These challenges highlight the need for more advanced approaches such as Retrieval-Augmented Generation (RAG) and knowledge graph integration, which aim to

improve factual accuracy, contextual grounding, and interpretability in generative systems.

2.7 Retrieval-augmented generation (RAG)

Retrieval-Augmented Generation (RAG) was introduced in 2020 by Patrick Lewis and a team of researchers at Facebook AI Research (FAIR), in collaboration with University College London and New York University. The paper addressed a key limitation of language models: their inability to access up-to-date or specific information outside their training data. RAG was introduced as a key approach for improving the reliability and factual grounding of generative AI systems. It combines information retrieval techniques with generative models to ensure that outputs are not only fluent but also grounded in external, verifiable knowledge sources (Lewis *et al*., 2020). In educational contexts, this is particularly important because accuracy and conceptual correctness are essential for effective learning (Kasneci *et al*., 2023).

The earliest implementation of RAG was termed naive RAG because it used sparse retrival techniques like BM25 to fetch documents based on keyword overlap. While efficient for simple factoid queries, it offered limited semantic understanding and often retrieved noisy content, failing to reason across multiple sources. By 2022, advanced RAG was developed, self-RAG for instance was introduced by Asai *et al* (2023). They added post retrieval stages, query rewriters to reformulate vague queries, and rerankers or refiners to reorder or compress noisy retrieved content before generation. These improvements were however still bolted onto a linear pipeline, the field needed a way to make the components self reconfigure for different tasks

In 2024 Gao *et al*. resolved the issue of linear pipeline with their discrete configurable modules, allowing developers to incorporate diverse techniques into specific stages. This improvement gave the field architectural flexibility, supporting both serialized pipeline execution and end to end learning across modular components. Modular pipelines were still largely reactive, retrieving once (or a fixed number of times) per

query, rather than reasoning about when and what to retrieve, or drawing on richer relational structure between places of knowledge (Gao *et al*. 2024).

Graph RAG and Agentic RAG introduced in 2024 resolved the issue of reactive pipelines through its multistep reasoning across knowledge sources and adaptive retrieval decisions, Graph RAG particularly brought in relational context between entities that flatterned document retrieval misses. Graph RAG is however still in its infancy stage experiencing several challenges which include its noisy graph construction, multimodal blindness and weak use of relational structure

2.7.2 Components of RAG systems

A standard RAG system is composed of several key components that work together to ensure effective information retrieval and generation.

The first component is the retriever, which is responsible for identifying relevant documents from a knowledge base using techniques such as dense vector search or semantic similarity matching (Karpukhin *et al*., 2020).

The second component is the knowledge base, which stores external information such as documents, articles, or structured datasets. This database serves as the source of factual grounding for the system.

The third component is the generator, typically a large language model, which uses the retrieved information to produce coherent and contextually relevant outputs (Lewis *et al*., 2020).

In some advanced systems, a reranker is also included to refine retrieved results and improve relevance before generation. Together, these components enable RAG systems to combine retrieval-based precision with generative flexibility. This architecture helps reduce dependency on memorized training data and allows models to access up-to-date and domain-specific information when generating responses.

2.7.3 Role of RAG in improving accuracy and context

One of the most important contributions of RAG is its ability to improve factual accuracy in generative AI systems. Traditional language models rely solely on learned parameters, which can lead to hallucinations or outdated information. By incorporating external retrieval, RAG reduces this limitation by grounding outputs in real data sources (Lewis *et al*., 2020).

A 2024 study by Pinecone tested RAG performance at scale using the Falcon RefinedWeb dataset, comprising 980 million web pages from commonCrawl, and found that RAG significantly improved LLM performance even on questions within the model's own training domain, with GPT-4's answer faithfulness improving by 13% once supplemented with RAG over the full billion document corpus, they also discovered that this positive effect increased with the amount of retrievable data made available, however, not every configuration produced games

RAG also enhances contextual relevance, as retrieved documents provide additional background information that helps the model generate more informed responses (Izacard & Grave, 2021). This is particularly useful in educational settings where explanations must be accurate, structured, and aligned with curriculum content.

Furthermore, RAG improves adaptability, allowing systems to incorporate new information without requiring full retraining of the underlying model. This makes it highly suitable for dynamic domains such as education, where knowledge frequently evolves (Kasneci *et al*., 2023).

In 2025 Wang *et al*. Carried out a study on the the construction and applicaton of retrieval enhanced generation (RAG) based on knowledge graph.they found that On the Natural Questions dataset, KG-RAG's ROUGE-L score improved from 41.2 to 46.9, its BLEU score rose from 31.5 to 38.7, and its FactScore increased by 13.6% compared to the original RAG model, On the PubMedQA medical question-answering task, KG-RAG achieved 81.3% accuracy — a 6.8 percentage-point improvement over

standard RAG demonstrating its advantage in knowledge reasoning within specialized fields, In the knowledge-enhanced dialogue task (OpenDialKG), KG-RAG achieved a BLEU-4 improvement of 4.8%, a ROUGE-L improvement of 5.5%, and a Knowledge F1 improvement of 11.5% over baseline models. The authors concluded that KG-RAG effectively addresses the shortcomings of traditional generative models

2.7.4 Applications of RAG in educational content generation

Following advances in retrieval-augmented generation and the significant reduction in hallucination it enabled, educational applications of RAG-based systems began to expand rapidly, particularly as researchers sought to address the hallucination and static internal knowledge that had limited the reliability of LLMs in educational settings (Li *et al*., 2025).

One of the earliest applied tests came from intelligent tutoring. Lee (2024) built a computer tutoring system that integrated generative AI with RAG to tailor instruction delivery and map responses to vetted educational materials. The experiment found that RAG-augmented systems performed far better than conventional static tutors by dynamically accessing applicable knowledge segments and anchoring LLM-synthesized answers in fact-based sources, reducing the risk of hallucinated answers while guaranteeing the pedagogical integrity of the system. By integrating domain-specific course content into vector databases, the tutor was able to contextualize student queries and deliver highly relevant instructional feedback.

Building on this, Liu *et al*. (2025) took a different approach with LPITutor, combining RAG with adaptive prompt engineering rather than retrieval alone. The study demonstrated that integrating RAG with prompt engineering improved response accuracy and relevance, leading to higher user satisfaction compared to standard LLM-based tutoring systems. This suggested that retrieval quality alone is not sufficient,

how retrieved content is presented to the model through prompt design also shapes output quality, a refinement Lee's earlier system did not test.

A related but architecturally distinct approach emerged with multi-agent, protocol-driven tutoring. Modran (2025) integrated RAG with a Model Context Protocol and Agent Communication Protocol to support adaptive tutoring in higher-education STEM courses, deploying it as a real classroom aid over several weeks of a Software for Telecommunications course. Usage logs showed several student interactions per week, and instructor monitoring confirmed the system's responses adhered to pedagogical rules such as offering hint-based coaching rather than direct answers, with student feedback measuring usability, perceived learning assistance, and overall satisfaction.

Most recently, researchers have begun testing whether structured, multimodal knowledge, not just retrieved text can further improve tutoring outputs. Deng and Yuan (2026) proposed an intelligent tutoring framework that automatically constructs multimodal knowledge graphs from both course videos and textbook PDFs, integrating tools such as FFmpeg, Whisper, and OCR to fuse auditory, visual, and textual knowledge before combining graph and vector retrieval to drive RAG-based question answering. This addresses the difficulty educational LLMs face in adapting to unstructured teaching resources and maintaining logical reasoning during question answering, though the system extracts knowledge from existing instructional video rather than generating new video content, leaving that direction untested.

2.8 Knowledge graphs in education

Knowledge Graphs (KGs) have become an important component in modern artificial intelligence systems due to their ability to represent knowledge in a structured and semantically meaningful way. In education, they are particularly valuable because learning content is inherently interconnected, requiring systems that can represent concepts, relationships, and dependencies clearly (Hogan *et al.*, 2021).

16 A Knowledge Graph is a structured representation of knowledge that models real-world entities and the relationships between them in a graph format. It typically consists of nodes (representing entities or concepts) and edges (representing relationships between those entities) (Hogan *et al*., 2021).

For example, in an educational context, “Photosynthesis” may be a node, while “requires sunlight” and “produces oxygen” may be relationships connecting it to other concepts. This structure allows information to be stored in a way that reflects how knowledge is naturally connected.

26 Knowledge graphs are widely used in search engines and recommendation systems because they support semantic understanding rather than simple keyword matching (Paulheim, 2017). These semantic relationships are often formally defined using ontologies, which provide a structured framework for organizing concepts and their interactions. In education, this makes them useful for organizing curriculum content in a logical and hierarchical manner.

2.8.1 Evolution of knowledge graphs

Knowledge graph development has progressed through four stages, each addressing limitations of the one before it.

68 In 2001, Berners Lee proposed the semantic web in an article co-authored with James Hendler and Ora Lassila, his argument was that the web, as it existed then, was built for human reading but not for machine undersanding, a webpage’s content is meaningful to a person looking at it, but a computer has no way to interpret what the words on that page actually mean or how they relate to one another. This idea led to the establishment of RDF and OWL as the technical foundation and produced manually curated resources like DBpedia and wordNet through rule-based, symbolic reasoning (Hogan *et al*., 2021). development prioritized formal correctness over scalability, limiting adoption to research contexts.

In 2012, google launched knowledge graphs, which proved that structured knowledge could operate at web scale (singhal, 2012), driving enterprise adoption for entity resolution, recommendation, and fraud detection (e.g., facebook's social graph, microsoft's satori). focus shifted from logical rigor towards practical, semi automatically constructed entity relationship stores.

By 2018, graph embedding techniques began to emerge, graph neural networks allowed knowledge graphs to feed directly into machine learning pipelines rather than being queried symbolically, enabling link predication and knowledge grounded Natural language processing. Knowledge graphs were further developed to converge with large language models via retrieval augmented generation extended over graph structures (Edge *et al*., 2024). the graph was used to ground the large language model output structurally while the LLM supplied fluent generation. This is termed the GraphRAG era. Existing graphRAG models however still face difficulty balancing retrieval accuracy against computational efficiency. Graph construction and traversal are inherently more expensive than flat vector search.

A congtrolled experiment carried out by Han *et al*. (2025) revealed that the two approaches show complementary behaviors rather than one consistently winning. standard RAG performed better on single hop, detail oriented questions, while GraphRAG performed better on multi hop, intensive reasoning questions. They also found that specific GraphRAG design choices matter, for instance, community-based global search can sacrifice query-specific detail, which affects detail-oriented tasks even while helping broader summarization tasks, and that evaluation protocol itself can shift conclusions, since LLM-as-a-Judge scoring for summarization proved sensitive to the order in which candidate summaries were presented

2.8.4 Applications in educational content organization

Knowledge graphs are increasingly used to organize educational content in a structured and meaningful way. One of the earliest concrete applications is automated assessment

generation. Chen and Shiu (2025) developed KAQG, a knowledge-graph-enhanced RAG pipeline that generates exam questions at controlled difficulty levels, using the graph's structure to reason about which concept combinations make a question harder or easier.

Building on this KAQG, Deng and Yuan (2026) went a step further to automate the construction of the knowledge graph itself directly from raw course material using FFmpeg for audio extraction, whisper for transcription, and OCR with layout analysis for textbook PDFs rather than relying on a pre-built or manually curated graph as previous systems did, allowing their hybrid graph and vector retrieval strategy to drive the tutoring question and answer. Their evaluation discovered that it performed better than both plain text RAG baseline and manually constructed knowledge graph basedline on answer accuracy, relevance and explainability, particularly on multi step reasoning questions.

More recently, knowledge graphs have been integrated into AI-based content generation systems, where they provide structured context for generating accurate and logically organized educational materials (Pan *et al*., 2023).

2.14 Research gaps

Despite significant advancements in Artificial Intelligence, Retrieval-Augmented Generation (RAG), and Knowledge Graphs (KGs), several important gaps still exist in the application of these technologies to educational video production systems. These gaps highlight the need for more integrated, context-aware, and pedagogically aligned solutions.

2.14.1 Lack of structured knowledge integration in video generation

Most existing AI-driven video generation systems rely heavily on unstructured data sources such as raw text, web documents, or pre-trained language models without explicit structured knowledge representation. Google NotebookLM for instance it

relies on standard Retrieval-Augmented Generation (RAG) using unstructured text embeddings, semantic search, and co-occurrence patterns to connect information across your uploaded documents.While these systems can generate fluent scripts, they often lack deep semantic organization of concepts (Hogan *et al*., 2021).

Knowledge Graphs, which provide structured relationships between concepts, are rarely integrated into video generation pipelines. As a result, generated educational videos may not follow logical learning sequences or properly represent prerequisite relationships between topics (Paulheim, 2017).

This lack of structured knowledge integration limits the pedagogical effectiveness of AI-generated educational videos, especially in complex subjects where concept dependency is important. these systems also lack systemic pedagogical constraints which indicates that they are not built native educational platforms built on a structured instructional framework (ACM DoCEng, 2025)

2.14.2 Limited context-aware educational content systems

Although RAG systems improve contextual relevance by retrieving external information, many educational applications still operate in a shallow retrieval mode that does not fully capture deeper instructional context (Lewis *et al*., 2020).

Existing systems often retrieve isolated documents without modeling the broader conceptual relationships between topics. This results in content that may be factually correct but not fully aligned with instructional flow or learning progression (Kasneci *et al*., 2023).

This limitation becomes more visible because videos require structured narration, smooth transitions, and logical sequencing of ideas, which are not always guaranteed by basic retrieval systems.

2.14.3 Need for improved accuracy and explainability

Another major gap in existing systems is the challenge of ensuring both accuracy and explainability in generated educational content. While generative models can produce high-quality outputs, they often lack transparency in how information is generated or retrieved (Bender *et al*., 2021).

Similarly, RAG systems improve factual grounding but do not always provide clear reasoning paths that explain why specific content was generated or selected. This reduces trust in AI-generated educational materials, particularly in formal learning environments where accountability is important (Ji *et al*., 2023).

Knowledge Graphs offer some level of explainability through explicit relationships between concepts, but they are often not fully utilized in generative pipelines. This creates a gap in systems that are both accurate and interpretable.

2.14.5 Limited use of mathematical and statistical validation frameworks

While graph-theoretic knowledge representation, similarity-based retrieval, and inferential statistical evaluation are individually well established, few studies apply these methods together within a single validation framework for AI-generated instructional videos. Existing evaluations of AI-generated educational content often rely on qualitative or descriptive assessments rather than formal statistical comparison against human-generated content (Leiker *et al*., 2023; Pellas, 2025). This represents a gap that the present study addresses by combining a graph-based knowledge representation, similarity-based retrieval, and a formal statistical comparison of learner outcomes within a single system.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

The method used for generating AI-based educational videos in this study is based on an integrated architecture combining a knowledge graph, a retrieval-augmented generation (RAG) component, and a pedagogical constraint layer. These components were selected based on the literature reviewed and the limitations identified in existing AI-generated instructional videos as discussed in chapter one, particularly their weak pedagogical structure, lack of organized prerequisite knowledge and tendency to produce inaccurate or fabricated content when relying solely on a language model's internal knowledge.

The model involves five (5) stages which include: construction of a knowledge graph from the uploaded course materials, indexing and retrieval of relevant content to ground generation in real source material, application of pedagogical constraints to structure each script segment, generation of narration audio from the constrained script, and generation of slides and assembly of the final instructional video.

3.2 Research design and rationale

This study was done following a design science research approach (Hevner et al. 2004; Peffers et al. 2007), in which a new product, KRAVE video generator in this case is designed, built and evaluated against criteria set in advance. Design science was adapted because the aim of the study is to produce and test a working system, not to test a theory. The work followed the procedures of design science by Peffers et al. (2007) which include problem identification and motivation, deginition of the objectives of the solution, design and development of the artefact, demonstration of the artefact on case studies and evaluation of the artefact. The results of the demonstration and evaluation are reported in chapter four, and discussed in chapter five.

3.3 System architecture

The system was organized as a pipeline of connected stages, where the output of one stage became the input of the next. Course materials supplied by the user, such as lecture notes or textbook content on a given topic, formed the starting point of the pipeline, and a finished instructional video was the final output. Figure 3.1 shows the stages of the pipeline.

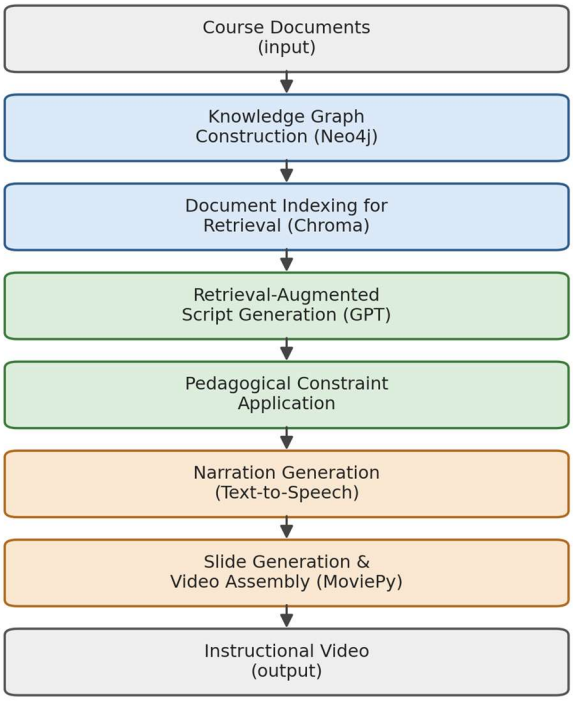


Figure 3.1. System architecture of the KRAVE video generator

Each stage of the pipeline was implemented as an independent module, so that documents could move through the pipeline step by step and any single stage could be modified, re-run, or tested on its own without affecting the others. Section 3.5 describes the implementation of each of these stages in detail.

3.4 Development tools and technologies

Table 3.1 summarizes the main tools and technologies used to implement the system.

Table 3.1. Development tools and technologies

Component	Tool / Technology	Purpose
Programming language	Python 3.10	Implementation of the entire pipeline
Language model	GPT-4o-mini (OpenAI large language model)	Concept extraction, content generation, and script writing
Knowledge graph storage	Neo4j	Optional Storage of extracted concepts and their relationships
Vector database	Chroma	Storing document embeddings for retrieval-augmented generation
Text-to-speech	GPT-based text-to-speech	Converting the generated script into narration audio
Image generation	gpt-image-2.5-flare	Generating images for slides in phase 4 (can be skipped with --no-images flag)
Video assembly	MoviePy	Stitching slides and narration into a final video file
Embedding model	openAi text-embedding-3-small, all-miniLML6-v2 of sentence transformers (local option)	Converting document chunks and concepts into vectors for similarity search
Development environment	Standard workstation with internet access	Running the pipeline and connecting to the language model API

3.5 System implementation

3.5.1 Knowledge graph construction

The knowledge graph module accepted the uploaded course documents and passed them to the GPT model with instructions to identify the key concepts discussed in the text and the relationships connecting them, such as prerequisite-of, part-of, or leads-to. The model returned these as concept-relationship-concept triples, which were then written into a Neo4j graph database using the Neo4j Python driver. Listing 3.1 shows a simplified version of this function. In the final implementation run through the runner script, this database write was switched off and the graph was kept as a JSON file (section 3.5.6); Listing 3.1 therefore shows the optional storage step.

Formally, the extracted concepts and their prerequisite relationships were treated as a directed graph

$$G = (V, E) \ldots. (3.1)$$

where V is the set of concepts extracted from the course materials and $E \subseteq V \times V$ is the set of prerequisite relationships between them. Because a segment could only be generated for a concept once its prerequisites had been covered, the order in which concepts were presented followed a topological ordering π of G, such that

$$\pi(a) < \pi(b) \text{ for every } (a, b) \in E \ldots\ldots\ldots (3.2)$$

for every prerequisite edge $(a, b) \in E$. This ordering constraint is what allowed the pedagogical constraint module (Section 3.5.3) to enforce scaffolding, so that a concept was never introduced before the concepts it depended on.

A topological ordering exists only if G contains no directed cycle. Cycle detection was therefore run on G before sequencing and before the critical analysis of section 3.5.3 the lowest confidence edge in the cycle was removed and recorded in the resolution plan.

Listing 3.1.: Knowledge graph extraction and storage (simplified)

```
def extract_knowledge_graph(document_text):
    """Send course text to GPT and return a list of
    (concept, relationship, concept) triples."""
    prompt = build_extraction_prompt(document_text)
    response = call_gpt(prompt)
    triples = parse_triples(response)
    return triples



def store_triples_in_neo4j(triples, driver):
    with driver.session() as session:
        for subject, relation, obj in triples:
            session.run(
                "MERGE (a:Concept {name: $subject}) "
                "MERGE (b:Concept {name: $object}) "
                "MERGE (a)-[r:RELATION {type: $relation}]->(b)",
                subject=subject, object=obj, relation=relation
            )
```

3.5.2 Content indexing and retrieval (RAG)

To reduce the risk of the language model generating inaccurate or fabricated information, commonly referred to as hallucination, the system did not rely solely on the language model's internal knowledge. The uploaded documents were split into smaller text chunks, converted into vector embeddings, and stored in a Chroma vector database. When content needed to be generated for a particular concept, the system retrieved the most relevant chunks from Chroma using similarity search and supplied them to the GPT model as supporting context, an approach widely known as retrieval augmented generation (Lewis *et al.* 2020). Listing 3.2 shows this retrieval step.

Relevance between a query and a stored chunk was measured using cosine similarity between their embedding vectors:

$$sim(q,d_i)=\frac{q.d_i}{|q||d_i|} \ldots\ldots (3.3)$$

138 where q is the embedding of the query and d_i is the embedding of a candidate chunk in the Chroma collection D. For each query, the top_k=4 highest-scoring chunks were retrieved and supplied to the language model as supporting context:

$$D^{*}=top{-}k\ sim(q,d_i) \ \ldots\ldots(3.4)$$

Generation was restricted to this retrieved set, rather than to the language model's unconstrained output. Chroma returns a distance rather than similarity score. The chroma collections were created with the default distance, squared Euclidean (L2) distance, and chroma returned this distance for each retrived chunk. Chunks were ranked in the same order by distance and by cosine similarity, so the top_k selection of Equation 3.4 was unaffected. A relevance floor was used as a threshold on this distance, and the corresponding cosine similarity is 0.25 which facilitated the restriction of generation to this retrieved set.

A further refinement was to introduce a relevance-bounded backfill step after the initial top_k retrieval. Let R denote the baseline set of top_k chunks retrieved by equation (3.4), and let S(c) denote the source document of chunk c. For each source documents not represented in R, its best-scoring candidate chunk was identified:

$$c^{*}_{s} = \text{argmax } \{ \text{sim}(q, c) : c \in D, S(c) = s \} \ldots\ldots..(3.5)$$

This candidate was admitted into the retrieved set if its score fell within a relaxation margin δ of the weakest chunk currently in R:

$$\text{sim}(q, c^{*}_{s}) \geq \tau - \delta, \text{ where } \tau = \min\{ \text{sim}(q, c) : c \in R \} \ (3.6)$$

Rather than fixing δ as a constant, it was defined relative to the score spread already present in R, so that the margin adapts to how decisively the top_k ranking separates one source from another:

$$\delta = \alpha \cdot (\text{sim}(q, c_1) - \tau), 0 < \alpha \leq 1 \ (3.7)$$

where c_1 is the highest-scoring chunk overall. A backfill cap of b_max additional chunks was imposed to bound the size of the final retrieved set, and the retrieved set size was allowed to grow elastically up to k + b_max rather than displacing already-retrieved chunks, so that no chunk scoring above τ was discarded to make room for a lower-scoring one. This preserved the guarantee that generation remained grounded in retrieved evidence the core hallucination-mitigation property described above while extending source coverage where genuinely warranted by relevance.

Listing 3.2.: Document indexing and retrieval for RAG

```
def index_documents(documents, chroma_collection):
    for doc_id, text in enumerate(documents):
        chunks = split_into_chunks(text, max_tokens=300)
        embeddings = embed_chunks(chunks)
        chroma_collection.add(
            ids=[f"{doc_id}-{i}" for i in range(len(chunks))],
            documents=chunks,
            embeddings=embeddings
        )



def retrieve_relevant_context(query, chroma_collection, top_k=4):
    results = chroma_collection.query(
        query_texts=[query],
        n_results=top_k
    )
    return results["documents"][0]
```

3.5.3 Pedagogical constraint application

The pedagogical constraint module was responsible for shaping the retrieved content into a form that supports learning rather than simply presenting information. The prerequisite ordering obtained from the knowledge graph was used to arrange concepts from simple to complex (scaffolding; wood *et al.* 1976), each script segment was

limited to a small number of new ideas (cognitive load theory; Sweller 1988), and every segment was required to follow a consistent structure of recap, new concept, example, and check for understanding (instructional sequencing). Listing 3.3 shows how these rules were combined into the prompt used to generate each segment.

Cognitive load theory was operationalised as a hard limit on the number of new concepts C_s introduced within a single script segment s:

$$|C_s| \leq N_{max} \text{ for every segment } s \quad \ldots\ldots. (3.8)$$

with N_max set to one in the prompt template shown in Listing 3.3, so that each segment introduced at most one new concept (exactly one in practice, since every segment had to teach its own concept) alongside a recap of the prerequisite concept, an example, and a check for understanding.

Listing 3.3.: Pedagogically constrained prompt template (simplified)

```
def build_segment_prompt(concept, retrieved_context, previous_concept):
    return f"""
    Write one short instructional video segment for the concept
    below, using ONLY the supporting context provided.

    Concept to teach: {concept}
    Previously covered concept: {previous_concept}
    Supporting context: {retrieved_context}

    Follow this structure:
    1. Brief recap of "{previous_concept}"
    2. Introduce "{concept}" in simple terms
    3. Give one short worked example
    4. End with a one-line check-for-understanding question

    Keep the segment focused on a single new idea only.
    """
```

Critical path analysis of the concept sequence

To show which concepts determine the length of the sequence, the critical path method (Kelley and Walker 1959) was applied to the directed acyclic graph of each case. Each concept v was given an estimated duration d(v) in minutes, from the word count of its script at the narration pace of 130 words per minute. The earliest start ES, earliest finish EF, latest start LS, latest finish LF and slack of each concept were then computed as:

$$ES(v) = \max\{ EF(u) : (u, v) \in E \}, \text{with } ES(v) = 0 \text{ if v has no prerequisite} \quad ...(3.8a)$$

$$EF(v) = ES(v) + d(v) \quad ..... (3.8b)$$

$$LF(v) = \min\{ LS(w) : (v, w) \in E \}, \text{ with } LF(v) = T \text{ if v has no successor, where } T = \max EF \quad ..... (3.8c)$$

$$LS(v) = LF(v) - d(v), \; slack(v) = LS(v) - ES(v) \quad ..... (3.8d)$$

Concepts with zero slack form the critical path: any delay to them delays the whole sequence. Where the concepts of a case are presented in a single prerequisite chain, every concept has zero slack and the critical path is simply the sum of all durations.

3.5.4 Narration Generation

Once a segment's script had passed through the pedagogical constraint module, it was sent to a GPT-based text-to-speech process to generate the narration audio for that segment. Listing 3.4 shows this step.

Listing 3.4.: Narration audio generation

```
def generate_narration(script_segment, output_path):
    audio = gpt_text_to_speech(text=script_segment, voice="default")
    with open(output_path, "wb") as f:
        f.write(audio)
    return output_path
```

3.5.5 Slide generation and video assembly

A slide was generated for each script segment to visually support the narration. MoviePy was then used to combine each slide with its matching narration audio clip, and simple transitions were added between segments, and finally the segments were rendered in sequence into a single instructional video file. Listing 3.5 shows a simplified version of the assembly step. In this study a segment is the script, slide and narration clip produced for one concept; a video is the file rendered for a concept or a group of concepts; and the output of a case study is the playlist of its videos, whose total duration is the sum of the durations of the videos.

Listing 3.5.: Slide and narration assembly into a video

```
from moviepy.editor import ImageClip, AudioFileClip, concatenate_videoclips


def assemble_video(slide_paths, audio_paths, output_path):
    clips = []
    for slide_path, audio_path in zip(slide_paths, audio_paths):
        audio_clip = AudioFileClip(audio_path)
        image_clip = ImageClip(slide_path).set_duration(audio_clip.duration)
        image_clip = image_clip.set_audio(audio_clip)
        clips.append(image_clip)

    final_video = concatenate_videoclips(clips, method="compose")
    final_video.write_videofile(output_path, fps=24)
    return output_path
```

3.5.6 Computer program implementation

The KRAVE pipeline described in Section 3.3 was implemented as a command-line program written in Python. The program is organized as four phases, each contained in a separate script, which are chained together by a runner script, run_pipeline.py. The

phases correspond to the stages of the pipeline as follows. Phase 1 builds the knowledge graph from the uploaded documents. Phase 2 embeds the document chunks and the knowledge graph concepts into a vector database. Phase 3 produces a personalized learning plan, applying the prerequisite ordering and pedagogical constraints and retrieving supporting chunks for every concept in the plan. Phase 4 renders the instructional videos from that plan. The script generator determines the number of slides and images based on the target duration, learning depth, Bloom's taxonomy level, and teaching mode. It generates slide titles, bullet points, narration, and image prompts, while incorporating supporting source chunks and transitions between concepts to ensure grounded explanations and continuity across videos. The resulting storyboards are validated and passed to subsequent modules for narration and slide assembly. Table 3.2 lists the script, input and output of each phase.

Table 3.2. Phases of the program, with their scripts, inputs and outputs

Phase	Script	Main input	Main output
1	kg/orchestrator.py	Source documents (a file or a directory)	Run folder containing knowledge_graph.json, extraction_report.json, resolution_plan.json and state.json
2	rag/rag_orchestrator.py	Phase 1 run folder	ChromaDB collections for chunks and concepts, grounding_index.json and rag_state.json
3	pedagogical/pedagogical_planner.py	knowledge_graph.json, a learner profile and the ChromaDB collection	learning_plan.json

Phase	Script	Main input	Main output
4	vidgen/video_orchestrator.py	learning_plan.json	One video for each concept or group of concepts in the plan, written to the output folder

The runner does not import the phase scripts. It launches each phase as a separate process using Python's subprocess module, shows its output as it runs, and stops with the failing exit code if any phase fails. The phases communicate only through files on disk, so any phase can be run again without repeating the earlier ones. After a phase finishes, the runner locates its output, namely the most recent run folder for Phase 1 and the most recent learning plan for Phase 3, and passes it to the next phase. A complete run is started with a single command that names the documents and the learner profile, for example python run_pipeline.py --docs ./documents/ --learner learner.json. Table 3.3 lists the main options of the runner.

Table 3.3. Main command-line options of the program runner

Option	Purpose
--docs	Source documents; starts the pipeline from Phase 1
--phase1-dir	Existing Phase 1 run folder; skips Phase 1
--plan	Existing learning plan; skips Phases 1 to 3
--start-phase, --end-phase	Run only a chosen range of phases
--learner	Learner profile file used in Phase 3
--no-rag, --no-openai	Skip retrieval grounding, or skip the language-model classification of concepts and use rules instead
--drop-optional	Remove optional concepts from the plan
--no-images	Skip image generation in Phase 4 and use plain backgrounds

Option	Purpose
--priority, --concepts	Render only concepts of a given priority, or only named concepts
--max-minutes, --total-minutes, --videos, --wpm	Limit the length of each video and of all videos combined, set a target number of videos, and set the narration pace in words per minute
--resume	Skip videos that have already been rendered

Phase 1 is run by a pipeline orchestrator that moves the documents through ingestion, extraction, resolution, review and an optional database write. Ingestion splits each document into chunks. Extraction sends each chunk to the language model (gpt-4o-mini in this case), in one or two passes, to obtain concepts and relationships, pausing between chunks and recording chunks that fail or return low-confidence results. Where several documents are supplied, their extraction reports are merged into one. Resolution then merges the extracted items into a single knowledge graph of nodes and edges and writes a resolution plan describing the changes made. A review queue of uncertain items can be produced, and the graph can be written to Neo4j through a separate writer module; when the program is started through the runner, the database write is switched off and the graph is kept as a JSON file. Settings such as the amount of delay between chunks, the confidence threshold and the output folder are held in a configuration object that is read from a YAML file or from environment variables, and credentials are read from the environment and not written in the code. The orchestrator saves a state file for every run, recording the run identifier, the source, the start and end times, the status, and the files produced. Each run is written to its own timestamped folder, so earlier runs are never overwritten. A run can be resumed from its state file, and a review-only mode reloads the artefacts of an earlier run and goes straight to the review and database stages.

Phase 2 loads the chunks from the extraction report and the concepts from the knowledge graph, skipping with a warning any chunk whose text was not saved, and

stops if no chunks remain. It then embeds the chunks and the concepts into two ChromaDB collections, stored in a persistent folder inside the run folder, in batches of 50 items. Embeddings are produced with OpenAI's text-embedding-3-small model by default, or with the all-MiniLM-L6-v2 model of the sentence-transformers library when the local option is chosen, which removes the need for an API key. The phase also writes a grounding index that links each concept to the chunks that support it, and a summary file recording the number of chunks and concepts embedded.

Phase 3 is run by a planner class that reads the learner profile, runs the sequencing engine, grounds the plan, and writes the result. The learner profile is a JSON file holding the learner's identifier and name, goals, the concepts already mastered, the concepts the learner finds difficult, the preferred learning style, and the depth of treatment. The sequencing engine takes this profile and the knowledge graph and returns an ordered plan in which mastered concepts are placed in a skipped list, missing prerequisites are inserted ahead of the concepts that need them, and each concept is given a priority (critical, standard or optional), a Bloom level (Bloom 1956), a teaching mode and a depth hint. Bloom levels are assigned by the language model, or by simple rules when the language model is switched off.

The plan is then grounded one concept at a time. The query is the concept's label together with its definition. More chunks than are needed are first fetched from ChromaDB, five times the number to be kept, and a second query restricted to chunks already linked to the concept is merged in, with duplicates removed. Chunks farther from the query than a distance threshold of 1.5 are discarded, and a source-diversity function, retrieve_diverse, then selects the final chunks so that no single source document dominates. The number of chunks kept for each concept (top_k) was 4. A concept is marked as grounded if at least half of top_k chunks were found, partially grounded if only some were found, and ungrounded if none were found. The finished plan is written as learning_plan.json, which holds a summary of the learner, the number of critical, standard and optional concepts, the prerequisite gaps found and the

concepts skipped, followed by every concept with its attributes, its retrieved chunks and its grounding status.

Source diversity was controlled by two rules. First, a source document qualified for a guaranteed slot only if its best chunk passed the relevance floor of 1.5 (the diversity-qualification floor). Second, each qualifying source was guaranteed at least [value, as in Table 4.1] chunk(s) in the final set (the minimum-per-source guarantee). The set then grew by at most b_max = [value] additional chunks through the backfill step of Equations 3.5 to 3.7, so the final set never exceeded k + b_max chunks.

Phase 4 receives the learning plan from the runner and renders the videos. Its options allow the run to be limited to one priority level or to named concepts, to be resumed after an interruption, to be restricted by the length of each video and of all videos combined, and to skip image generation so that plain coloured backgrounds are used instead of generated images. The narration pace, which defaults to 130 words per minute, links the length of a script to the duration of its video. Because one video is produced for each concept or group of concepts, the output for a case study is a playlist of videos.

The main libraries used were the Anthropic and OpenAI client libraries for the language models, ChromaDB for vector storage, the Neo4j driver for the optional database write, sentence-transformers for local embeddings, NumPy and scikit-learn for numerical work, PyYAML for configuration, and python-dotenv for reading credentials. Every phase was built to write timestamped log messages, and every phase could be run on its own from the command line, which allowed each stage to be tested separately as described in Section 3.7.1.

3.6 Case studies

The system was applied to three case studies, each drawn from a different subject area, to show that it can be used on content of different kinds and not only on a single topic. The first case study, Product Development, was drawn from an Industrial Engineering course. The second was quadratic equations from ordinary level mathematics and the

third was the Rule of Law from ordinary level Government. For each subject, course materials were uploaded into the system, which constructed a knowledge graph, indexed the materials for retrieval, applied the pedagogical constraints, and generated a complete instructional video playlist covering the selected sub-topics of that subject.

The subjects were chosen to differ in the nature of their content, for example, while one course was theory based, another was calculation based. The diversity in the nature of concepts was meant to test the scaffolding and instructional sequencing features. Course materials of sufficient length also had to be available for each subject. Table 3.4 lists the case studies.

Table 3.4. Case studies used to evaluate the system

Case	Subject area	Source course	Nature of content
1	Product Development	Industrial Engineering	conceptual and process-based
2	Quadratic equations	O'level mathematics	calculation-based
3	Rule of Law	O'level Government	Theory based

3.7 System testing

Testing of the system took place at two levels: functional testing of the pipeline itself, and an evaluation of the generated videos involving learners and evaluators. The procedures followed for both are described below; the results obtained are presented in Chapter Four.

3.7.1 Functional testing of the pipeline

Each stage of the pipeline, knowledge graph construction, retrieval, pedagogical constraint application, narration generation, and video assembly, was tested individually using sample course documents before the stages were connected together. This was done to confirm that each stage produced a correctly formatted

output that the next stage could accept, for example, that extracted triples were in a form the Neo4j storage function could consume, and that generated scripts were in a form the narration function could accept. The complete pipeline was then tested end-to-end using the materials of the case studies, to confirm that a finished video could be produced from uploaded documents without manual intervention between stages.

In addition, each pedagogical and retrieval constraint described in this chapter was checked against the generated output, for example that concepts followed prerequisite order and that each segment introduced a single new concept, and the processing time of each stage was recorded. The criteria applied and the results obtained are reported in Section 4.2.

3.7.2 Evaluation design

To address Objective 5 in Chapter One, the system was evaluated through the videos it generated for the three case studies. No human-made comparison video was used. Instead, each video was judged against standards set in advance, so that the evaluation shows whether the system meets defined standards of learning and instructional quality. A comparison with human-made videos was outside the scope of this study and is recommended as further work.

The evaluation had two strands. Learners were examined through a pretest to test their knowledge of the topic. Following that, learners watched the video for a case and then took a second quiz, and the learning gain was measured. The Learning outcomes and the quality of instructional design were checked using both using both learners and instructors, and because three subjects were used, results were analyzed within each case and across cases, to show whether the system performed consistently across subjects.

3.7.3 Participants and evaluators

Learners were students who had not previously studied the selected sub-topics in depth. Each case study had its own group of learners, and each learner was assigned to

one case study only, so that the groups are independent as required by the one-way analysis of variance, Levene's test and the benchmark tests. with some learners taking more than one case to see their learning gain across different domains. A total of 30 learners took part. They were SS1 students selected at random from a pool of students from two schools (The international school Ibadan and Baptist college university of Ibadan). The videos were rated by 5 evaluators, who were subject experts with experience in the relevant subjects and in teaching. Each evaluator received the template and a short briefing on the rating scale and rated the videos independently, without discussing scores with other evaluators.

Sample size. Thirty learners were recruited for each case study, giving 90 learners in total. This target was based on a power analysis for the one-sided one-sample t-test against the mastery benchmark and for the one-sided paired t-test of learning gain (α = 0.05, power = 0.80). Twenty-seven learners per case are needed to detect a medium effect (d = 0.5) and 12 to detect a large effect (d = 0.8) (Cohen 1988). A sample of 30 per case therefore exceeds the requirement for a medium effect and allows for learners who withdraw or fail to complete the tests. Because each learner took one case only, the three groups are independent. With 30 learners per group, the one-way ANOVA across cases has power of about 0.90 to detect a large difference between cases (f = 0.40) but only about 0.50 for a medium difference (f = 0.25), so a non-significant result should not be read as evidence that the cases performed equally. Thirty learners per case also gives KR-20, the Shapiro–Wilk test and the exact confidence intervals reasonable stability.

3.7.4 Evaluation instruments

Three instruments were used: a pre-test and a post-video quiz for each case, and the Educational Video Instructional Quality Evaluation Template. The pre-test and the post test used the same items.

A quiz of 15 multiple-choice items was prepared for each case to test comprehension and retention of the concepts covered in the video, and was reviewed by a subject

expert for relevance to the content. Each item was mapped to one sub-topic of the case in a quiz blueprint (Appendix C), so that performance could be reported by sub-topic. It was administered immediately after learners finished watching the video, and each item was marked correct or incorrect. The internal consistency of each quiz was measured with the Kuder–Richardson formula 20 (Kuder and Richardson 1937), which is the form of Cronbach's alpha for items scored in this way:

$$KR{-}20 = \frac{k}{k-1}\left(1 - \frac{\sum_{i=1}^{k} p_i q_i}{\sigma_X^2}\right) \quad \ldots\ldots\ldots(3.9)$$

where k is the number of items, p_i is the proportion of learners who answered item i correctly, $q_i = 1 - p_i$, and σ_x^2 is the variance of the learners' total scores. A value of 0.70 or above was taken as acceptable. Each item was also described by its difficulty, the proportion of learners who answered it correctly, and its discrimination, the difference between the proportion who answered it correctly in the highest-scoring and lowest-scoring [27]% of learners. Item difficulty between 0.30 and 0.70 and discrimination of at least 0.30 were taken as acceptable (Ebel and Frisbie 1991).

Instructional quality was assessed with the Educational Video Instructional Quality Evaluation Template (Appendix B), which contains 34 criteria grouped into ten dimensions (A to J), listed in Table 3.5.

Table 3.5. Dimensions of the evaluation template

Code	Dimension	Number of criteria
A	Learning objectives and alignment	3
B	Accuracy and subject content	3
C	Instructional design and pedagogy	4
D	Clarity and explanation	4
E	Learner engagement	3
F	Visual and audio quality	4
G	Accessibility and inclusiveness	4

Code	Dimension	Number of criteria
H	Assessment and learning check	3
I	Relevance and practical application	3
J	Overall instructional effectiveness	3
	Total	34

Each criterion was rated from 1 (Poor) to 5 (Excellent). A video's total score T is the sum of its 34 ratings and has a maximum of 170, and its average rating A is the total divided by 34:

$$A=\frac{1}{34}\sum_{j=1}^{34} r_j \quad \ldots\ldots\ldots\ldots(3.10)$$

where r_j is the rating given to criterion j. The average rating was interpreted with the bands in Table 3.6, and the template also records a recommended action for the video based on this rating. The printed total of 150 on the original form was corrected to 170 to match its 34 criteria, and a Poor band (1.00 to 1.49) was added so that the bands cover the full scale of 1 to 5.

Table 3.6. Interpretation of the average rating

Average rating	Quality level	Interpretation
4.50 – 5.00	Excellent	Strong instructional resource; suitable for use with minimal or no revision
3.50 – 4.49	Good	Effective overall; minor improvements may strengthen learning
2.50 – 3.49	Satisfactory	Usable with targeted improvements to instructional quality
1.50 – 2.49	Needs Improvement	Significant revisions recommended before use
1.00 – 1.49	Poor	Not suitable for use in its present form

3.7.5 Data analysis procedure

Data were analysed with Microshof Excel and Python (3.11, numpy, pandas and scipy,), and a significance level of $\alpha = 0.05$ was adopted throughout, so that a result was considered statistically significant where $p < 0.05$. The results are presented in Chapter Four and interpreted in Chapter Five.

Quiz scores were converted to percentages so that cases with different quizzes could be compared, and were summarised for each case and for all cases together. The Shapiro–Wilk test (Shapiro and Wilk 1965) was used to check whether the scores in each case were approximately normally distributed, and Levene's test (Levene 1960) was used to check for equal variances across cases. For each case, a one-sample t-test (H_1: mean score > B) examined whether the mean score was above the mastery benchmark B:

$$t = \frac{\bar{X} - B}{s/\sqrt{n}} \quad \ldots\ldots\ldots (3.11)$$

where $\bar{X}$ is the mean score of the case, s is the standard deviation of the scores and n is the number of learners. The size of the difference between the mean and the benchmark was reported using Cohen's d (Cohen 1988), with 0.2, 0.5 and 0.8 taken as small, medium and large:

$$d = \frac{\bar{X} - B}{s} \quad \ldots.(3.12)$$

The one-sample Wilcoxon signed-rank test (Wilcoxon 1945), which does not assume normality, was used as a robustness check. Because one test was made for each case, the p-values were adjusted with the Holm procedure (Holm 1979) to control the chance of a false positive across the tests. The percentage of learners who reached B was also reported with an exact (Clopper-Pearson) 95% confidence interval (Clopper and Person 1934), since a mean above the benchmark can hide a group of learners below it. Differences in performance between the cases were tested with a one-way analysis of variance, with the Kruskal–Wallis test (Kruskal and Wallis 1952) as a check that does not assume normality. The effect size was η^2 for the analysis of variance and ε^2 for the Kruskal–Wallis test. Where either test was significant, pairwise comparisons were

made with Holm adjustment [state the post-hoc test used, e.g. Games–Howell or Dunn's test]. Because some cases had few learners (Case 2 had N = 6), the quiz reliability, normality tests and confidence intervals were interpreted with caution for those cases, and the non-parametric checks were given priority. Learning gain was examined by comparing the pre-test and post-test percentage scores of each case with a one-sided paired t-test (H_1: post-test mean > pre-test mean), with Holm adjustment across the cases. The paired effect size d_z was reported, being the mean gain divided by the standard deviation of the gains, together with the normalised gain g (Hake 1998), which is the share of the possible improvement that was achieved:

$$g=\frac{\bar{X}_{\text{post}}-\bar{X}_{\text{pre}}}{100-\bar{X}_{\text{pre}}} \quad \text{.....(3.13)}$$

where $\bar{X}$_post and $\bar{X}$_pre are the mean post-test and pre-test scores of the case, in percent.

Evaluator ratings were summarised as the mean and standard deviation of each dimension and as the total score and average rating of each video. To keep observations independent, each evaluator's ratings were first averaged across the cases, so that each evaluator contributed one score per dimension. The one-sample Wilcoxon signed-rank test then examined, one sided (Hi: median > 3.50), whether each dimension's ratings were above 3.50, with Holm adjustment across the ten dimensions and the effect size r reported (0.1, 0.3 and 0.5 taken as small, medium and large). Friedman's test (Friedman 1937) was used to examine whether evaluators' total scores differed between cases.

The reliability of the ratings was checked in three ways. Internal consistency of the 34 criteria was measured with Cronbach's alpha (Cronbach 1951) and McDonald's omega (McDonald 1999), where alpha is

$$\alpha=\frac{k}{k-1}\left(1-\frac{\sum_{i=1}^{k}\sigma_i^2}{\sigma_T^2}\right) \quad \text{.....(3.14)}$$

with k the number of criteria, σ_i^2 the variance of ratings on criterion i, and σ_t^2 the variance of the total scores. Agreement between evaluators was measured with the

17 intraclass correlation coefficient ICC(2,k) (Shrout and Fleiss 1979), and agreement between pairs of evaluators on the 1 to 5 rating given to each criterion of each video with quadratic weighted kappa, averaged over evaluator pairs. Kappa was not computed on the quality level of each video, because with only three videos, the levels five too few observations and most videos fall in the same band. Values of 0.70 or above were taken as acceptable for alpha and omega, and 0.60 or above as moderate or better for the ICC and kappa.

3.7.6 Ethical considerations

105 119 The learners were SS1 students and therefore minors.from the department, a written note was sent to the concerned schools to obtain consent for the study. Participation was voluntary and learners could withdraw at any time without penalty. Scores were recorded under anonymous codes with no names, and the data were stored securely and used only for this study. Evaluators took part voluntarily and gave their ratings anonymously.

CHAPTER FOUR

RESULTS

4.1 Introduction

This chapter presents the results obtained from testing the KRAVE system described in Chapter Three. The results are reported in the same order in which the system was built and evaluated. Section 4.2 reports the functional testing of the pipeline, including the checks made on the pedagogical constraints. Section 4.3 describes the output produced when the pipeline was applied to the 2 case studies, which cover different subject areas. Section 4.4 reports the evaluation of the KRAVE videos for each case study and across all of them, first through learner performance on the post-video quiz and then through ratings of instructional quality. Section 4.5 summarizes the chapter.

Results are presented in tables, and each table is followed by a short account of what it shows and how it should be read. The interpretation of the results in relation to the objectives of the study, and the reasons behind them, are left to Chapter Five. All statistical tests were carried out at the significance level of $\alpha = 0.05$ adopted in Section 3.7.5, using Microsoft excel and python.

4.2 Functional Testing Results

4.2.1 Test configuration

Before any result is reported, Table 4.1 records the settings under which the pipeline was run. Reporting these values makes the tests repeatable and allows the later results to be traced to a specific configuration of the system.

Table 4.1: Configuration of the pipeline during testing

Parameter	Value used
Language model (concept extraction and script writing)	Gpt-4o-mini
Chunk size for indexing	300 tokens (maximum)
Number of chunks retrieved per query (top_k)	4
RAG relevance floor (maximum distance)	1.5

Parameter	Value used
Minimum chunks guaranteed per qualifying source	50
Maximum new concepts per segment (N_max)	1
Optional-content exclusion (--drop-optional)	on
Learner profile (learning style, depth)	intermediate

4.2.2 Stage-level and end-to-end results

Each stage of the pipeline was first tested individually against an acceptance criterion drawn from the design in Chapter Three. The full pipeline was then run end-to-end on the materials of each case study without manual intervention between stages. Table 4.2 lists the criterion applied to each stage and the outcome. A stage is recorded as passed only when its output met the criterion without manual correction.

Table 4.2: Functional testing results for each pipeline stage

Pipeline stage	Acceptance criterion	Remark
Knowledge graph construction	Concepts and prerequisite relationships extracted and stored; graph passes the directed acyclic graph (cycle-detection) check	Passed: Concepts and prerequisite relationships were extracted and stored successfully. Cycle detection confirmed that the graph was acyclic.
Content indexing and retrieval	Retrieved chunks fall within the relevance floor; top_k = 4 respected; every qualifying source contributes at least its guaranteed minimum	Passed: Relevant content chunks were retrieved within the specified relevance threshold. The top_k = 4 limit and minimum source contribution requirements were satisfied

Pipeline stage	Acceptance criterion	Remark
Pedagogical constraint application	Concepts follow prerequisite order; each segment introduces one new concept and follows the recap–concept–example–check structure	Passed: All concepts followed the prerequisite order, and each segment adhered to the recap–concept–example–check structure without manual correction.
Narration generation	An audio file is produced for every script segment	Passed: Audio narration was generated successfully for every script segment, with no manual intervention required.
Slide generation and video assembly	One slide per segment is combined with its narration into a single playable video file	Passed: Each script segment was matched with its corresponding slide and narration, and the resulting video played successfully.
End-to-end pipeline	A complete video is produced from the uploaded documents with no manual step between stages	Passed: The complete pipeline produced a playable instructional video from the uploaded materials without manual intervention between stages.

Of the six tests, all passed at the first attempt. Although the narration generation initially produced too brief explanations, this was as a result of a hard coded word limit in the script_generator.py script. A later modification which allowed video duration to be set as an argument resolved this issue.

4.2.3 Verification of the pedagogical constraints

The constraints described in Section 3.3.3 are the mechanism through which the pipeline enforces instructional structure, so each was verified separately. Table 4.3 states the check applied to each constraint and whether the generated output satisfied

it. A constraint is recorded as satisfied only if no violation was found in any case study output.

Table 4.3: Verification of the pedagogical and retrieval constraints

Constraint	Verification check	Violations found	Outcome
Prerequisite precedence	Compared the order of concepts in the plan with a topological ordering of the knowledge graph	0	Satisfied
Cycle handling	Ran cycle detection on the graph before the critical path method	0	Satisfied
Mastery exclusion	Confirmed that concepts marked as mastered were placed in the skipped list	0	Satisfied
Prerequisite-completeness repair	Checked that every missing prerequisite was inserted ahead of the concept that needs it	0	Satisfied
RAG relevance floor	Checked that no retrieved chunk lay beyond the distance limit of 1.5	0	Satisfied
Diversity-qualification floor	Checked that only sources whose best chunk cleared the floor received a guaranteed slot	0	Satisfied
Minimum-per-source guarantee	Counted chunks per qualifying source against the configured minimum	0	Satisfied
Retrieval budget cap	Confirmed that the total chunks returned never exceeded k	0	Satisfied
Optional-content exclusion	Confirmed that optional concepts were dropped only when the flag was on	0	Satisfied
Learner profile validity	Attempted to construct a profile with a value outside the allowed set	0	Satisfied

4.2.4 Processing time

Table 4.4 reports the mean time taken by each stage across the case studies, measured on dell latitude 5520 core i5 11th generation 16gb ram. These figures show where the pipeline spends its time and whether a complete video can be produced within a practical period.

Table 4.4: Mean processing time per pipeline stage (per case study)

Stage	Time (min:s)	Share of total (%)
Knowledge graph construction	2:10	11.7
Content indexing and retrieval	0:10	0.9
Script generation under constraints	3:20	18.0
Narration generation	2:04	11.2
Slide generation and video assembly	10:45	58.2
Total	18:29	100

Most of the time was spent during the slide generation and video assembly session, accounting for 58 percent of the overal time, the total time however stayed within a window of 18 minutes 29 seconds

4.3 Output of the Case Studies

The pipeline was applied to 2 case studies drawn from different subject areas, as described in Section 3.6. Using more than one subject shows whether the pipeline performs consistently across content of different kinds, rather than only on the topic for which it was first developed. This section reports what the pipeline produced at each stage for every case, starting with the knowledge graphs and ending with the generated videos.

4.3.1 Overview of the case studies

Table 4.5 summarises the case studies and the size of the knowledge graph and video produced for each. Differences in the number of concepts and segments reflect differences in the amount and structure of the source material, and are reported so that the later comparison of results across subjects can be read in context.

Table 4.5: Overview of the case studies and their generated outputs

Case	Subject area	Docs	Concepts	Edges	Segments	Duration
1	Product Development (Industrial Engineering)	2	176	210	100	132:40

Case	Subject area	Docs	Concepts	Edges	Segments	Duration
2	Quadratic equations (Ordinary level science)	1	64	130	7	42:10
3	Rule of Law	1	36	100	9	45

Table 4.6 shows how the relationships extracted from each case were distributed by type. The proportion of prerequisite relationships matters because that relationship drives the sequencing of concepts.

Table 4.6: Relationships in the knowledge graphs by type and case

Relationship type	Case 1	Case 2	Case 3	Total
prerequisite-of				
part-of				
leads-to				
Other				
Total				

4.3.2 Concept sequencing and critical path

Concepts were ordered by prerequisite precedence, and the critical path method of Section 3.3.2 was applied to the directed acyclic graph of each case. Table 4.7 gives the full analysis for Case 1 as an illustration, using the estimated duration d of each concept in minutes. Concepts with zero slack form the critical path: any delay to them delays the whole sequence. The equivalent tables for the other cases are given in Appendix 4.1.

Table 4.7: Critical path analysis of the Case 1 (introduction to quadratic equations) concepts (minutes)

Concept	d	ES	EF	LS	LF	Slack	Critical path

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Concept	d	ES	EF	LS	LF	Slack	Critical path
Quadratic equations	[X]	[X]	[X]	[X]	[X]	[X]	[Yes / No]
Factorization method	[X]	[X]	[X]	[X]	[X]	[X]	[Yes / No]
Quadractic formula methods	[X]	[X]	[X]	[X]	[X]	[X]	[Yes / No]
Completing the square method	[X]	[X]	[X]	[X]	[X]	[X]	[Yes / No]
Factor form of graph	[X]	[X]	[X]	[X]	[X]	[X]	[Yes / No]
Common mistakes in quadractics	[X]	[X]	[X]	[X]	[X]	[X]	[Yes / No]
Word problems							

8 strategy

In Case 1 the critical path consisted of 7 concepts as highlighted above, giving a total sequence duration of 42 minutes. Across the [K] cases the critical path contained between [XX] and [XX] concepts, and the total sequence duration ranged from [XX] to [XX] minutes.

4.3.3 Retrieval grounding

Table 4.8 shows how the retrieved chunks were distributed across the source documents of each case. Because the source-diversity strategy of Section 3.3.1 is designed to stop one document from dominating retrieval, the share column is the main evidence of whether it worked.

Table 4.8: Distribution of retrieved chunks across source documents

Case	Source document	Chunks indexed	Chunks retrieved	Share of case (%)
1	Quadratic equations			
2	Product development			

Case	Source document	Chunks indexed	Chunks retrieved	Share of case (%)
3	Rule of Law			

4.3.4 Generated videos

The pipeline generated one video for each case study. Table 4.9 lists the segments of the Case 1 video as an illustration, with each segment's concept, its duration, and whether it contained all four required elements of the segment structure. The same check was applied to every segment of every case, and the proportion of segments with the complete structure is stated below the table.

Table 4.9: Structure of the Case 1 video by segment

Segment	Concept	Duration (s)	Recap	Concept	Example	Check
1	Quadratic equations	3:14	[✓]	[✓]	[✓]	[✓]
2	Factorization method	2:27	[✓]	[✓]	[✓]	[✓]
3	Quadractic formula methods	2:52	[✓]	[✓]	[✓]	[✓]
4	Completing the square method	2:50	[✓]	[✓]	[✓]	[✓]
5	Factor form of graph	[XX]	[✓]	[✓]	[✓]	[✓]
6	Common mistakes in quadractics	[XX]	[✓]	[✓]	[✓]	[✓]
7	Word problems strategy					

Add one row per segment. Mark a missing element with a dash.

Across all [K] cases, 7 of 7 segments (100%) contained all four elements, and the concept order followed the knowledge graph in all cases. No segment needed manual review and the case it belonged to.

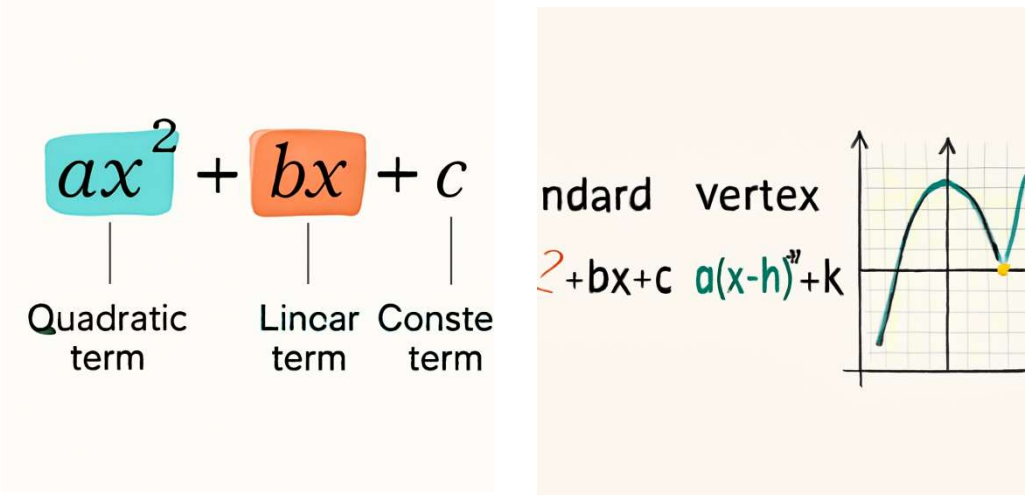


Figure 4.1 A sample slides and script except from the first concept of case 1

4.4 Evaluation Results

This section reports the evaluation of the KRAVE videos described in Section 3.7.2. Several concept videos and were generated for each of the 3 case studies, and each video was evaluated in two ways. Learner performance was measured with a post-video quiz for each case, and analysed by comparing scores against a pre-specified mastery benchmark and across cases. Instructional quality was measured by evaluators who rated each video on a structured template, and was analysed by comparing ratings against the threshold for a good rating and across cases. Because every case was evaluated in the same way, the results show not only whether the videos met these standards but also whether they did so consistently across subjects. Using both kinds of evidence allows the result to be checked from the learners' side and from the evaluators' side.

4.4.1 Participants and evaluators

Table 4.10 gives the number of learners who watched each video and the number of evaluators who rated it. Learners had not studied the subject in depth before watching, which is what allows their quiz performance to be linked to the video, so the level of study is reported alongside the numbers.

Table 4.10: Learners and evaluators by case study

Case	Learners (N)	Level of study	Evaluators (N)
1. Product Development	2	—	2
2. Quadratic Equations	2	SS1	2
3. Rule of Law	3	SS1	3
Total	7		7

A total of 7 learners took part across the 3 cases. The learners were selected at random from a pool of students who had not studied the subject before. Each learner watched the videos for a particular case study once and took the assessment. The evaluators were selected from experienced teachers on the subject matter.

4.4.2 Quiz instruments

A quiz was prepared for each case and given to learners immediately after they watched the video. Because every conclusion about learner performance depends on the quizzes measuring comprehension consistently, the quality of each was checked first. Each item was marked correct or incorrect, so internal consistency was measured with the Kuder–Richardson formula 20 (KR-20). Item difficulty is the proportion of learners who answered an item correctly, and item discrimination is the difference in that proportion between the highest- and lowest-scoring learners. Table 4.11 gives the results.

Table 4.11: Quality indicators of the post-video quizzes

Case	Items	KR-20	Mean item difficulty	Mean item discrimination
1. Product Development	12	Not computed	0.83	Not computed
2. Quadratic Equations	12	Not computed	0.71	Not computed
3. Rule of Law	12	Not computed	0.86	Not computed

Guideline

4.4.3 Learner performance

Because the quizzes differ between cases, scores are expressed as percentages. Table 4.12 presents the descriptive statistics for each case and for all cases together. The mean shows the typical score, the standard deviation shows how widely scores were spread, and the median and range show whether a few extreme scores affected the mean.

Table 4.12: Descriptive statistics of quiz scores (%) by case

Case	N	Mean	Standard deviation	Median	Min – Max
1. Product Development	2	83.33	11.79	83.33	75.00 – 91.67
2. Quadratic Equations	2	70.83	17.68	70.83	58.33 – 83.33
3. Rule of Law	3	86.11	4.81	83.33	83.33 – 91.67
All cases	7	80.95	11.50	83.33	58.33 – 91.67

Because there was no comparison video, the scores were judged against a mastery benchmark (B) of 60%, set in advance as described in Section 3.7.5. The question asked of each case was whether learners who watched the KRAVE videos scored, on average, above this benchmark. Before the tests were run, their assumptions were checked, as shown in Table 4.13. The Shapiro–Wilk test examines whether the scores in each case are approximately normally distributed, and Levene's test examines whether the cases have similar variances, which matters for the comparison across cases. A p-value above 0.05 on either test means that no departure from the assumption was detected.

Table 4.13: Assumption checks for the analysis of quiz scores

Test	Statistic	p-value	Assumption met
Shapiro–Wilk, Case 1	Not computable (n = 2)	—	—

Test	Statistic	p-value	Assumption met
Shapiro–Wilk, Case 2	Not computable (n = 2)	—	—
Shapiro–Wilk, Case 3	W = 0.750	< 0.001	No
Levene's test across cases	F(2, 4) = 36.79	0.003	No

Table 4.14 reports the comparison with the benchmark for each case. A one-sample t-test examines whether the mean score is above B, and the p-values were adjusted with the Holm method because one test was made for each case and the chance of a false positive grows with the number of tests. Cohen's d, here the difference between the mean and the benchmark in standard deviation units, shows how far above or below B the typical learner scored, with 0.2, 0.5 and 0.8 conventionally taken as small, medium and large. The last column gives the percentage of individual learners who reached B, with an exact 95% confidence interval, since a mean above B can hide a sizeable group of learners below it. The one-sample Wilcoxon signed-rank test, which does not assume normality, was used as a robustness check and agreed in direction with the t-test in every case, but with two or three learners per case its smallest attainable one-sided p-value is 0.25 and 0.125 respectively, so it could not reach significance in any individual case (pooled across the seven learners, p = 0.016).

Table 4.14: Quiz scores compared with the mastery benchmark in each case

Case	Mean (%)	t (df)	p (Holm-adjusted)	Cohen's d vs B	Reached B (%) (95% CI)
1	83.33	2.80 (1)	0.218	1.98	100.0 (15.8–100.0)
2	70.83	0.87 (1)	0.273	0.61	50.0 (1.3–98.7)
3	86.11	9.40 (2)	0.017	5.43	100.0 (29.2–100.0)
All	80.95	4.82 (6)	0.001	1.82	85.7 (42.1–99.6)

Whether performance differed between subjects was tested with a one-way analysis of variance (ANOVA) across the cases, with the Kruskal–Wallis test as a check that does not assume normality. Table 4.15 gives the results; where the test was significant, pairwise comparisons with Holm adjustment are reported below the table.

Table 4.15: Comparison of quiz performance across the case studies

Test	Statistic	p-value	Effect size
One-way ANOVA	$F(2, 4) = 1.19$	0.393	$\eta^2 = 0.37$
Kruskal–Wallis test	$H(2) = 1.58$	0.454	$\varepsilon^2 = 0.26$

Table 4.16 compares pre-test and post-test scores with a paired t-test, reports the paired effect size d_2 (the mean gain divided by the standard deviation of the gains), and gives the normalised gain, which is the share of the possible improvement that was achieved.

Table 4.16: Pre-test and post-test scores (%) by case

Case	Pre-test mean	Post-test mean	Mean gain	Paired t (df), p (Holm)	d_2	Normalised gain
1	No pre-test given	—	—	—	—	—
2	No pre-test given	—	—	—	—	—

4.4.4 Performance by sub-topic

To show which parts of the content the videos taught well and which they taught less well, the quiz items were grouped by sub-topic. Table 4.17 gives the percentage of correct answers for each sub-topic, and whether it reached the benchmark. A sub-topic that falls below B points to content that the video explained less effectively, which is useful for improving the pipeline.

Table 4.17: Quiz performance per sub-topic for rule of law (% correct)

Case	Sub-topic	Items	% correct	Reached B
1	Understanding law and order	—	—	—
1	Why society needs laws	—	—	—
2	Definition of rule of law	—	—	—
2	Dicey's Three Principles	—	—	—
3	Features of a society that respects rule of law	—	—	—
3	Fair hearing and fundamental human rights	—	—	—

Item-level responses were not recorded in this round of testing, so quiz performance could not be broken down by sub-topic.

4.4.5 Instructional quality ratings

Instructional quality was assessed with the Educational Video Instructional Quality Evaluation Template. The template contains 34 criteria grouped into ten dimensions (A to J): learning objectives and alignment; accuracy and subject content; instructional design and pedagogy; clarity and explanation; learner engagement; visual and audio quality; accessibility and inclusiveness; assessment and learning check; relevance and practical application; and overall instructional effectiveness. Each criterion is rated from 1 (Poor) to 5 (Excellent). A video's total score is the sum of its 34 ratings (maximum 170), and its average rating is the total divided by 34. The average is interpreted with the bands in Table 4.18.

Table 4.18: Interpretation of the average rating

Average rating	Quality level	Interpretation
4.50 – 5.00	Excellent	Strong instructional resource; suitable for use with minimal or no revision

Average rating	Quality level	Interpretation
3.50 – 4.49	Good	Effective overall; minor improvements may strengthen learning
2.50 – 3.49	Satisfactory	Usable with targeted improvements to instructional quality
1.50 – 2.49	Needs Improvement	Significant revisions recommended before use

source: Educational Video Instructional Quality Evaluation Template; the Poor band

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Table 4.19 gives the mean rating and standard deviation of each dimension, averaged over all evaluators, over all case studies, and over the criteria within the dimension, together with the percentage of criteria rated 4 (Good) or above. The mean shows how the videos fared on each dimension, and the percentage shows how many individual criteria reached a good rating.

Table 4.19: Mean rating and standard deviation per dimension (scale 1–5)

Dimension (number of criteria)	Mean	SD	Rated 4 or above (%)
A. Learning objectives and alignment (3)	4.31	0.58	84.2
B. Accuracy and subject content (3)	4.10	0.92	72.2
C. Instructional design and pedagogy (4)	3.89	0.79	75.0
D. Clarity and explanation (4)	3.88	0.93	59.3
E. Learner engagement (3)	3.52	0.77	57.1
F. Visual and audio quality (4)	4.29	0.93	78.6
G. Accessibility and inclusiveness (4)	3.98	0.97	66.7
H. Assessment and learning check (3)	3.38	0.78	47.6
I. Relevance and practical application (3)	3.71	0.78	61.9
J. Overall instructional effectiveness (3)	4.10	0.94	71.4

Since there was no comparison video, each dimension was tested against a fixed standard: the lower limit of the Good band, 3.50. For each dimension, each evaluator's ratings were first averaged across the case studies, so that every evaluator contributes

one score and the observations are independent. The one-sample Wilcoxon signed-rank test then examined whether these scores were above 3.50. The test does not assume normality and suits ordinal ratings from a small number of evaluators. Because ten dimensions were tested, the p-values were adjusted with the Holm method. Table 4.20 reports the tests, with the effect size r (the standardized test statistic divided by the square root of the number of observations; 0.1, 0.3 and 0.5 are conventionally small, medium and large).

Table 4.20: Dimension ratings tested against the Good threshold (3.50)

Dimension	Z	Effect size r	p (Holm-adjusted)	Above 3.50
A. Learning objectives and alignment	2.23	0.84	0.130	No
B. Accuracy and subject content	1.62	0.61	0.421	No
C. Instructional design and pedagogy	1.27	0.48	0.506	No
D. Clarity and explanation	1.19	0.45	0.506	No
E. Learner engagement	0.53	0.20	0.745	No
F. Visual and audio quality	1.79	0.68	0.330	No
G. Accessibility and inclusiveness	1.58	0.60	0.421	No
H. Assessment and learning check	-0.43	-0.16	0.745	No
I. Relevance and practical application	0.68	0.26	0.745	No
J. Overall instructional effectiveness	1.44	0.54	0.448	No

Table 4.21 narrows the view to individual criteria by listing the five highest- and five lowest-rated criteria, averaged across cases and evaluators. This shows exactly where the videos were judged strongest and weakest, which the dimension averages can hide.

Table 4.21: Highest- and lowest-rated criteria

Rank	Criterion	Mean rating
1 (highest)	The video uses inclusive, respectful, and culturally appropriate examples.	4.86
2 (highest)	Video quality, transitions, and pacing support learning.	4.71

Rank	Criterion	Mean rating
3 (highest)	On-screen elements are not overcrowded or distracting.	4.57
4 (highest)	The video focuses on appropriate and achievable learning goals.	4.50
5 (highest)	The video could be used independently or effectively within a lesson.	4.43
1 (lowest)	The video includes questions, checks for understanding, or opportunities for practice where appropriate.	2.86
2 (lowest)	Questions, activities, scenarios, demonstrations, or other engaging elements are used effectively.	2.86
3 (lowest)	Learners receive useful feedback or guidance where applicable.	3.14
4 (lowest)	The content is usable by learners with diverse needs where applicable.	3.33
5 (lowest)	The presenter explains concepts clearly and at an appropriate pace.	3.43

Table 4.22 gives the mean total score each evaluator assigned across the case studies. It lets the reader see whether the overall result reflects the view of all evaluators or only a few.

Table 4.22: Mean total score awarded by each evaluator across cases

Evaluator	Mean total score (of 170)	Mean average rating
Evaluator 1	111.0	3.26
Evaluator 2	152.5	4.48
Evaluator 3	136.0	4.00
Evaluator 4	109.9	3.23
Evaluator 5	154.5	4.55
Evaluator 6	156.0	4.59
Evaluator 7	114.4	3.36
All evaluators	133.5	3.93

Because quality may depend on the subject, Table 4.23 gives the total score, average rating and quality level of the video in each case. Each evaluator rated the video of one case only, so the ratings are independent across cases and Friedman's test, which needs related ratings, could not be used. The Kruskal–Wallis test, which does not assume normality, was used instead to test whether evaluators' total scores differed between cases: $H(2) = 0.61$, $p = 0.738$.

Table 4.23: Total score, average rating and quality level by case

Case	Total (of 170)	Average rating	Quality level
1. Product Development	132.2	3.89	Good
2. Quadratic Equations	135.2	3.98	Good
3 Rule of Law	133.2	3.92	Good

Overall scores, taken as the mean across the case studies, are given in Table 4.24.

Table 4.24: Overall instructional quality score (mean across cases)

Measure	Value
Total score (out of 170)	133.5
Average rating (out of 5.00)	3.93
Quality level (Table 4.18)	Good
Criteria rated 4 or above (%)	67.5
Recommended action (from template)	Effective overall; minor improvements may strengthen learning

4.4.6 Reliability of the ratings

The ratings are only meaningful if the evaluators applied the template consistently, so three aspects of reliability were checked, summarised in Table 4.25. Cronbach's alpha and McDonald's omega were used to measure how consistently the 34 criteria behave as one scale; omega is reported alongside because the criterion did not contributes

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equally. The intraclass correlation coefficient, in its two-way random-effects, absolute-agreement, average-measures form ICC(2,k), measures how closely the evaluators agreed with each other. Weighted kappa measures agreement between evaluators on the quality level assigned to each video, giving credit for near agreement. Values of 0.70 or above are conventionally taken as acceptable for alpha and omega, and values of 0.60 or above are taken as moderate or better for ICC and kappa.

Table 4.25: Reliability statistics for the evaluation template

Statistic	Value	95% confidence interval	Threshold met
Cronbach's alpha (34 criteria)	0.92	0.81 – 0.98	Yes
McDonald's omega	Not estimable	—	—
ICC(2,k) between evaluators	0.02 – 0.39 (within case)	—	No
Weighted kappa on quality level	Not computable	—	—
Range of corrected item-total correlations	0.06 – 0.91	—	—

Note: Seven of the 238 ratings (2.9%) were missing and were left out of the means and percentages; for the reliability statistics each was replaced by the mean of the other ratings for that criterion. Because each evaluator rated the video of one case only, ICC(2,k) was computed within each case (Rule of Law 0.02, Quadratic Equations 0.29, Product Development 0.39) and its range is reported. McDonald's omega could not be estimated, because only seven raters were available for 34 criteria, and weighted kappa could not be computed, because no two evaluators rated the same video on the quality level of a video.

4.4.7 Summary of the statistical results

Table 4.26 highlights the overall results, what was tested, with which method, and what was found. Where the learner-based and evaluator-based results agree, the finding is stronger than either alone.

Table 4.26: Summary of the statistical tests applied in the evaluation

Question	Method	Main result	Decision

Question	Method	Main result	Decision
Did learners score above the mastery benchmark?	One-sample t-test with Holm adjustment; Wilcoxon check	Pooled mean 81.0% against B = 60%: t(6) = 4.82, p = 0.001, d = 1.82; 85.7% of learners reached B. By case, only Rule of Law was significant after Holm adjustment (p = 0.017)	Benchmark exceeded overall; significant for Rule of Law only
Did performance differ by subject?	One-way ANOVA; Kruskal–Wallis	ANOVA F(2, 4) = 1.19, p = 0.393; Kruskal–Wallis H(2) = 1.58, p = 0.454	No significant difference
Did learners improve after the video?	No pre-test was given	Not assessed	—
Were ratings above the Good threshold?	One-sample Wilcoxon with Holm adjustment	Overall mean rating 3.93; no dimension was significantly above 3.50 after Holm adjustment (smallest adjusted p = 0.130)	Not demonstrated at α = 0.05
Did ratings differ by subject?	Kruskal–Wallis test (Friedman not applicable: each evaluator rated one video)	H(2) = 0.61, p = 0.738	No significant difference
Was the quiz reliable?	KR-20	KR-20 not computed (item-level responses not recorded)	Not assessed

Question	Method	Main result	Decision
Were the ratings reliable?	Alpha, omega, ICC, weighted kappa	α = 0.92; ICC(2,k) within case 0.02 – 0.39	Scale consistent; inter-rater agreement below threshold

4.5 Summary

This chapter reported the results of testing the KRAVE video generation system. The functional test reported no bug, except for when the artificial intelligent (AI) application programming interface (API) was unreachable, the KRAVE pipeline was designed to auto resume however, so a break didn't affect the flow, instead, already completed concepts were skipped, all six test in Table 4.2 were passed and all ten constraints were satisfied. The 3 case studies produced 3 videos with a total of [XX] segments, built from knowledge graphs of [XX] to [XX] concepts. Learners who watched the videos scored a mean of 81.0% against a mastery benchmark of 60%, with 85.7% of learners reaching the benchmark, and performance did not differ significantly between subjects. Evaluators gave the videos a mean template rating of 3.93, which falls in the Good band. The reliability checks indicate that the rating scale was internally consistent (α = 0.92) but agreement between evaluators within each case was low, and quiz reliability could not be assessed because item-level responses were not recorded. The next chapter discusses these results in relation to the objectives of the study, and presents the conclusion and recommendations.

CHAPTER FIVE

DISCUSSION, CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

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This chapter interprets the results reported in Chapter Four in relation to the objectives of the study. Section 5.2 summarises the main findings. Section 5.3 discusses them, taking in turn the functional behaviour of the pipeline, the effect of the pedagogical constraints, the performance of learners, the instructional quality of the videos and the reliability of the evidence. Section 5.4 states the limitations of the study, Section 5.5 gives the conclusion, Section 5.6 states the contributions of the work, and Sections 5.7 and 5.8 give recommendations and suggestions for further research.

The evaluation reported in Chapter Four was small. Seven learners took part, distributed over three case studies as two learners for Product Development, two for Quadratic Equations and three for Rule of Law. The interpretations below are therefore stated as preliminary evidence, and the limits of that evidence are set out in Section 5.4.

5.2 Summary of Findings

The findings of Chapter Four can be summarised as follows.

1. **Functional performance.** All six pipeline stages passed their acceptance criteria, and the full pipeline produced a playable video from uploaded documents without manual intervention between stages. All ten pedagogical and retrieval constraints were satisfied, with no violation found in any case study output.

2. **Processing time.** A complete run took a mean of 18 minutes 29 seconds per case study on a Dell Latitude 5520 (Core i5, 11th generation, 16 GB RAM). Slide generation and video assembly took the largest share, 58.2% of the total.

3. **Case study outputs.** The pipeline was applied to three subject areas: Product Development (Industrial Engineering), Quadratic Equations (Ordinary level

mathematics) and Rule of Law (Ordinary level Government). It produced one video for each, from knowledge graphs of very different sizes.

4. **Learner performance.** The seven learners had a mean post-video quiz score of 80.95% (SD = 11.50), against a mastery benchmark of 60%. Six of the seven learners (85.7%) reached the benchmark. The pooled mean was significantly above the benchmark ($t(6) = 4.82$, $p = 0.001$, $d = 1.82$). By case, the means were 83.33% for Product Development, 70.83% for Quadratic Equations and 86.11% for Rule of Law, and only Rule of Law was significantly above the benchmark after Holm adjustment ($p = 0.017$). Performance did not differ significantly between cases (ANOVA $p = 0.393$; Kruskal–Wallis $p = 0.454$).

5. **Instructional quality.** The videos had a mean template rating of 3.93 out of 5 (a total of 133.5 out of 170), which is in the Good band, and 67.5% of individual ratings were 4 or above. The ratings were similar across cases (3.89, 3.98 and 3.92), and the difference was not significant (Kruskal–Wallis $p = 0.738$). No dimension was significantly above the Good threshold of 3.50 after Holm adjustment.

6. **Reliability.** The 34 criteria of the rating template behaved consistently as a scale (Cronbach's alpha = 0.92, 95% CI 0.81 to 0.98), but agreement between evaluators who rated the same video was low (ICC(2,k) between 0.02 and 0.39 within cases).

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5.3 Discussion of Results

5.3.1 Functional behaviour of the pipeline

The functional tests show that the stages of the pipeline can be chained into a working system: knowledge graph construction, content indexing and retrieval, constraint application, narration, and slide and video assembly each met their acceptance criteria, and the end-to-end run needed no manual step. This matters because the central claim of the study is that a language model, retrieval-augmented generation and a knowledge graph work better together than any of them alone. A system that fails at the hand-over

between stages cannot test that claim, and the first-attempt pass of all six tests suggests that the interfaces between the stages were well specified.

The one problem found during testing was that narration was too brief. It was traced to a fixed word limit in the script generator rather than to a weakness in the design, and it was resolved by making the target video duration an input to the pipeline. This is a useful result in itself. It shows that the length and depth of the narration are parameters of the system and not fixed properties of the method, so the pipeline can be adjusted for lessons of different length. The design of the pipeline to resume after an interruption of the language model service, skipping concepts already completed, also meant that an unreachable service delayed a run without corrupting it.

5.3.2 Effect of the pedagogical and retrieval constraints

All ten constraints were satisfied with zero violations. In practical terms, concepts were always taught after their prerequisites, mastered concepts were skipped, missing prerequisites were inserted, and no retrieved passage lay beyond the relevance limit. These are the properties that distinguish the pipeline from a general-purpose tool that asks a language model to write a lesson script, because in the latter case the order of ideas is left to the model and is not checked.

There are two points of caution. First, a constraint being satisfied shows that the pipeline did what it was designed to do. It does not show by itself that the design choice improves learning, which would need a comparison against a version without the constraints. Second, the verification was carried out on the outputs of three cases only. The zero-violation result is consistent with correct implementation, but the number of cases is too small to rule out violations in unusual source documents.

The structural check of the videos points the same way. Every segment was designed to follow the recap, concept, example and check structure, and the case videos were built around one new concept per segment. This one-concept-at-a-time structure is consistent with the aim of limiting the load placed on a learner at any moment, although the present study did not measure cognitive load directly.

5.3.3 Processing time

Most of the processing time went to slide generation and video assembly (58.2%), followed by script generation under constraints (18.0%), knowledge graph construction (11.7%) and narration (11.2%). Indexing and retrieval were almost negligible (0.9%). The practical meaning is that the intelligent parts of the pipeline, which are the knowledge graph, retrieval and the constrained scripting, are not the bottleneck. The time is spent in rendering, which is a mechanical step and the most likely to improve with better hardware or a more efficient rendering routine. A complete video in under twenty minutes on a standard laptop is a reasonable cost for a teacher, particularly compared with the hours usually needed to plan, record and edit a lesson video by hand, although this study did not time the manual alternative and the comparison is therefore only indicative.

5.3.4 Learner performance

The learners who watched the videos scored well above the mastery benchmark on average, and six of the seven reached it. The effect size for the pooled result is large ($d = 1.82$). This is encouraging, since the learners had not studied the subjects before and learned from the video alone.

The result must be read with its limits in mind. With seven learners, the pooled test has little power and the confidence interval for the share reaching the benchmark is wide (42.1% to 99.6%). The case-level results show the same thing. Rule of Law was significantly above the benchmark, with a very large effect size, but this partly reflects how closely the three learners' scores agreed (83.33% to 91.67%), which made the standard deviation small and the effect size large. Product Development had a mean of 83.33% but was not significant, because with two learners the test has almost no power to detect a difference. The Wilcoxon signed-rank check was in the same direction as the t-test throughout, but with two or three learners its smallest possible p-value is 0.25 and 0.125 respectively, so it could not reach significance in any single case. This is a limitation of the sample and says nothing about the videos.

Quadratic Equations had the lowest mean (70.83%), and one of its two learners scored 58.33%, which was the only score below the benchmark. This is the one result that deserves attention, because quadratic equations is a procedural topic in which learners must carry out calculations, whereas Rule of Law is largely conceptual. A video that explains and shows worked examples may be enough for conceptual material but less effective for procedural skills that need practice. The difference between cases was not statistically significant, and with one learner below the benchmark in a group of two it would be wrong to conclude that the pipeline is weaker on calculation-based subjects. It is, however, a hypothesis that a larger study should test directly.

5.3.5 Instructional quality

The mean template rating of 3.93 places the videos in the Good band, meaning that they were judged effective overall with room for minor improvement. About two thirds of the individual ratings were 4 or above. Eight of the ten dimensions had means above 3.50, but none was significantly above it after the p-values were adjusted for ten tests. With seven evaluators the test cannot reach significance for a small adjusted p-value, and the smallest adjusted p-value (0.130, for learning objectives and alignment) should be seen in that light. The reasonable reading is that the ratings lean towards Good, and that this lean is not yet statistically established.

The pattern across criteria is informative. The highest-rated criteria were the use of inclusive and respectful examples (4.86), the quality, transitions and pacing of the video (4.71), uncluttered on-screen elements (4.57), achievable learning goals (4.50) and suitability for independent use (4.43). These reflect strengths of the design. The slide generation step produces clean, readable slides, and the knowledge graph keeps each segment focused on a small number of goals, which is consistent with the high ratings for learning objectives (4.31) and for visual and audio quality (4.29).

The lowest-rated criteria were the inclusion of questions or checks for understanding (2.86), the effective use of engaging activities and scenarios (2.86), useful feedback to learners (3.14), usability for learners with diverse needs (3.33) and a clear explanation at an appropriate pace (3.43). The two dimensions with the lowest means were

assessment and learning checks (3.38) and learner engagement (3.52). These are linked. Although every segment of the pipeline includes a check, a check in a recorded video is a narrated prompt and the learner cannot answer it and receive feedback. Evaluators therefore judged it as weak, even though it is present in the structure. This is a limit of the video format and suggests that the check stage should be paired with an interactive element, such as a short quiz after each segment.

The ratings did not differ significantly between the three subjects, with mean ratings of 3.89, 3.98 and 3.92. This is consistent with the claim that the pipeline gives similar quality across subjects, but the evaluation covers only three subjects and seven evaluators, so it supports consistency only weakly.

The learner and evaluator evidence agree in direction. Learners scored above the benchmark and evaluators rated the videos in the Good band. The two sources also point to the same weakness, since the one learner below the benchmark and the low ratings for checks and practice both relate to opportunities for learners to practise. Where independent sources agree, the finding is stronger than either alone, although both sources here are small.

5.3.6 Reliability of the evidence

76 The 34 criteria behaved consistently as a single scale (alpha = 0.92), and the corrected item-total correlations ranged from 0.06 to 0.91, which means that a few criteria correlated weakly with the rest and may be measuring something different. The scale is therefore internally consistent as a whole.

Agreement between evaluators was low. Within each case the intraclass correlation was between 0.02 and 0.39, below the 0.60 level taken as moderate. The evaluator totals ranged from 109.9 to 156.0 out of 170, so some evaluators were consistently stricter than others. Three evaluators gave mean ratings below 3.4, while the other four gave mean ratings of 4.0 or higher. This spread is the main reason why the dimension

154 tests were not significant, and it shows that the quality level of the videos depends on who is rating. It would be misleading to report the mean rating without this variation.

Two further points affect how the reliability figures can be read. McDonald's omega could not be estimated because there were only seven raters for 34 criteria, and weighted kappa could not be computed because each evaluator rated one video only. The quiz reliability (KR-20) and item discrimination could not be computed because the answers to individual questions were not recorded. The quality of the quizzes as measuring instruments has therefore not yet been demonstrated, and the learner results depend on the assumption that the quizzes measured comprehension of the video content.

5.4 Limitations of the Study

The findings should be read with the following limitations in mind.

1. **Small sample.** Only seven learners took part, and only two or three learners were in each case. The tests have low power, confidence intervals are wide, and effect sizes are unstable. The results are preliminary evidence of feasibility and not a basis for generalisation.
2. **No comparison condition.** No human-made video or non-video lesson was used for comparison, so the study cannot show that the generated videos are better than, or equal to, other ways of teaching the same content. Learner scores were compared with a fixed benchmark and not with a control group.
3. **No pre-test.** Because no pre-test was given, learning gain could not be measured. The learners were selected as having not studied the subject, but a high post-test score is not the same as a demonstrated gain.
4. **Missing item-level data.** Without the answers to individual questions, quiz reliability, item discrimination and performance by sub-topic could not be reported.
5. **Rater variation and evaluator design.** Evaluators differed widely in how strictly they rated, and each evaluated one video only, so the differences between cases and between evaluators cannot be separated.

6. **Missing ratings.** Seven of the 238 ratings (2.9%) were missing. They were left out of means and percentages and replaced by the criterion mean in the reliability analysis, which slightly affects those statistics.

7. **Scope of subjects and learners.** The three cases cover two Ordinary level school subjects and one engineering topic, and the learners came from a single setting. The results may not hold for other subjects, levels, languages or learner groups.

8. **Dependence on external services.** The pipeline relies on a commercial language model service. Its outputs may change with model updates, and a repeat of the study might not produce identical videos.

9. **Short-term measurement.** Learners took the quiz immediately after viewing. Retention and transfer were not measured.

5.5 Conclusion

The study set out to build and evaluate a system, named KRAVE, that generates educational videos by combining a language model, retrieval-augmented generation and a knowledge graph under pedagogical constraints. The system was built as a four-stage pipeline and tested on three subjects.

The evidence supports four conclusions. First, the pipeline works as a whole: every stage passed its test, the end-to-end run needed no manual intervention, and a complete video was produced in under twenty minutes on a standard laptop. Second, the pedagogical and retrieval constraints are enforced reliably, because none of the ten constraints was violated in any case. Third, learners who watched the videos scored above the mastery benchmark on average, and most of them reached it, although only the Rule of Law result was individually significant and the sample was very small. Fourth, the videos were rated in the Good band for instructional quality, with strengths in inclusiveness, visual and audio quality and focus, and weaknesses in checks for understanding, engagement and feedback.

The results do not show that KRAVE videos are better than videos made by teachers, and they do not show a measured gain in learning. They do show that the approach is

technically sound and that it produces videos of acceptable instructional quality that learners can learn from. This is enough to justify a larger evaluation, and the weaknesses identified give a specific direction for the next stage of development.

5.6 Contributions of the Study

The study makes the following contributions.

1. **An integrated pipeline.** A working pipeline that connects knowledge graph construction, retrieval-augmented generation, constrained sequencing and video production, so that a teacher can go from a set of documents to a complete instructional video.
2. **Prerequisite-aware sequencing.** The use of the knowledge graph, with cycle detection and the critical path method, to order concepts by their prerequisites, and to skip mastered concepts and repair missing prerequisites.
3. **Source-diversity-aware retrieval.** A retrieval strategy that stops one source document from dominating the retrieved content when several documents define the same concept, while keeping a relevance limit on what is retrieved.
4. **An evaluation design.** A two-sided evaluation that combines learner performance against a pre-set mastery benchmark with ratings of instructional quality on a structured 34-criterion template, together with reliability checks for both.
5. **Evidence of feasibility.** Early evidence across three subject areas that the approach produces videos of Good instructional quality that learners can use to reach the mastery benchmark.

5.7 Recommendations

On the basis of the findings, the following recommendations are made.

1. **Add interactive checks.** The check stage of each segment should be paired with an interactive question and immediate feedback, for example a short quiz after each

segment. This addresses the lowest-rated criteria and the dimensions of assessment and engagement.

2. **Add practice for procedural topics.** For calculation-based subjects such as quadratic equations, the pipeline should add worked examples that the learner is asked to attempt, or a set of practice problems with answers, because watching a procedure is not the same as being able to carry it out.
3. **Improve accessibility.** Captions or transcripts, and narration that explains visual content, should be generated automatically, since usability for learners with diverse needs and the explanation of visual information were among the lower-rated criteria.
4. **Allow pacing to be set.** The video duration, narration depth and speaking pace should be options available to the teacher, since evaluators rated pace and length lower than other aspects of delivery.
5. **Keep teachers involved.** Teachers should review the knowledge graph and the script before the video is produced, because evaluator ratings of accuracy and level of detail varied, and a review is the simplest way to catch errors.
6. **Record item-level data in future evaluations.** The online examination system should store the answer to each question for every learner so that reliability, item discrimination and sub-topic performance can be reported.
7. **Train evaluators before rating.** Evaluators should be given a short calibration session with sample videos before using the template, to reduce the differences in strictness observed in this study.
8. **Optimise rendering.** Because slide generation and video assembly take most of the processing time, the rendering step should be the first target for efficiency improvements.

5.8 Suggestions for Further Research

The following areas are suggested for further work.

1. **A larger evaluation.** Repeat the evaluation with enough learners and evaluators per subject to give adequate statistical power, and report the results by subject with confidence intervals.
2. **A controlled comparison.** Compare KRAVE videos with videos made by teachers and with ordinary classroom instruction on the same content, using a pre-test and a post-test so that learning gain can be measured.
3. **Retention and transfer.** Test learners again after an interval and with new problems, to find out whether what is learned from the videos lasts and transfers.
4. **Procedural subjects.** Investigate whether performance is lower for calculation-based subjects than for conceptual ones, and whether adding practice activities removes the difference.
5. **Ablation studies.** Test the contribution of each component by running the pipeline with and without the knowledge graph, the source-diversity retrieval and the pedagogical constraints.
6. **Personalisation.** Evaluate the learner profile options of the pipeline, including learning style and depth, to find out whether adapting videos to a learner improves outcomes.
7. **Other languages and contexts.** Extend the pipeline to local languages and to other levels of education, and test whether its quality holds for different learner groups.
8. **Learner experience.** Collect learners' own views on clarity, engagement and workload, to complement the quiz scores and evaluator ratings.

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